2020

Advanced MTO Process in India

<u>Group #10</u>
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Project Charter

Project Title	Team Members
Advanced MTO Process in India	Sinan Masoud Khalil Shehadeh Emily Sobie Esteban Vazquez-Elizondo
Goals	**Deliverables**
• Analyze, understand, and accurately describe the processes used to produce olefins • Illustrate the optimal process design for olefin production • Design MTO simulation on Aspen • Develop Mass Balances for products • Hand Calculations and utilities in process • Process Control • Cost Analysis • Environmental Analysis	• Design basis and plan • Market analysis • Block flow diagram • Process flow diagram • Mass balance • Hand calculations • Aspen Simulation • Utility Table • Cost Calculations • HAZOP Analysis
In Scope	**Out of Scope**
• Safety Protocol • Cost Analysis • Process Control	• Optimization of existing processes • Delivery or distribution of product

Design Plan

Advanced MTO Process					
Task ID	Task	Start Date	Duration (Days)	Finish Date	Assignee
Logistics		14-Jan	3	17-Jan	All
1	First Meeting (Task Delegation)	14-Jan	1	15-Jan	All
2	Market Analysis Research	14-Jan	3	17-Jan	Khalil
3	Technology Comparison Research	14-Jan	1	17-Jan	Sinan
4	BFD & PFD Research	14-Jan	1	17-Jan	Emily, Esteban
5	Project Charter	14-Jan	1	15-Jan	Esteban
6	Design Plan	14-Jan	1	15-Jan	Esteban
15	Layout/Format	14-Jan	1	15-Jan	Esteban
Execution		17-Jan	7	28-Jan	All
Introduction		17-Jan	7	28-Jan	Khalil
7	Objective	17-Jan	7	28-Jan	Khalil
8	Market Analysis	17-Jan	7	28-Jan	Khalil
Process Description		17-Jan	7	28-Jan	Esteban, Khalil
9	Problem	17-Jan	7	28-Jan	Khalil, Esteban
10	Possible Solutions	17-Jan	7	28-Jan	Sinan
11	Chosen Solution	17-Jan	7	28-Jan	Esteban
12	Design Basis and Assumptions	17-Jan	7	28-Jan	Emily
13	BFD & Rough Mass Balance	17-Jan	7	28-Jan	Emily
14	PFD & Explanation	17-Jan	7	28-Jan	Esteban
Wrap-Up		28-Jan	2	30-Jan	All
16	Section Unification	28-Jan	1	29-Jan	All
17	Proofreading / Plagiarism Check	29-Jan	1	30-Jan	All
Appendix		20-Feb	9	29-Feb	All
9	Final PFD(Visio)	21-Feb	7	28-Feb	Esteban
10	Final BFD and Hand Calculations	21- Feb	7	28-Feb	Emily
11	Aspen Simulation	21- Feb	7	28-Feb	Esteban, Khalil
12	Mass Balances	21- Feb	7	28-Feb	Esteban, Khalil
13	Equipment List	21- Feb	7	28-Feb	Esteban, Khalil
14	Utilities	21- Feb	7	28-Feb	Sinan
Wrap-Up		27- Feb	2	29-Feb	All
16	Section Unification	28- Feb	1	29-Feb	All
17	Proofreading / Plagiarism Check	29- Feb	1	29-Feb	All

Introduction:

Clear Definition of the Problem

MTO is one of the most important processes in the world to date. MTO, which stands for methanol-to-olefins, provides the link between the use of coal and natural gas to the petrochemical industry. This process is difficult to do depending on the location of the plant and the type of resources available. The cost of raw materials in that location is another factor to consider. Coal and natural gas are used to produce methanol and then the MTO process begins where the methanol is converted to olefins. Ethylene and propylene olefins are produced, and plastics, polymers, and fibers can be made from the resulting product. The big problem in this process is in the demand for ethylene and propylene globally and specifically in the location of the plant in India. There is a 4% increase in demand globally with an 8-10% increase in demand for ethylene and propylene, specifically in India.[1] As of today, India is producing about 9 million metric tons of ethylene and propylene annually. Furthermore, there is an increase in demand of about 600,000 metric tons of ethylene and propylene per year in India.[2] This increase can lead to the supply in India not meeting the demand of ethylene and propylene. Furthermore, India has approximately a population of 1.38 billion people with an annual growth rate in population of about 1.01%.[3] This means that the population increases by approximately 14 million people born per year, which explains the reason behind the increase in demand of this product. There is a growth in middle class from the past 5-10 years in India which will lead to more people affording cars and buying more product. India's current economy is in the top 10 nation worldwide with soon to be in the top 5. Another problem that is emerging is that there is a lack of investment in undeveloped countries which have the resources that can help meet these demands. By investing in these underdeveloped countries, their economies have the potential to be boosted enough to meet the economic standards that developed countries have set. A lack of investment in these countries creates a rise in poverty, lack of resources, and a poor economy in these countries.

Why this problem is important?

Considering the global demand for ethylene and propylene, it is necessary to ensure that the global and the Indian market are satisfied. Having a growing population in a country that is not able to satisfy the needs of its population such as fuel, plastic production, or providing enough jobs can lead to its citizens migrating to other countries. India is currently dependent on natural gas and oil imports. The reason for this is due to the coal quality being poor in comparison to other countries. Since Africa has an abundant source of natural gas, then it will be able to meet the demand due to petrochemical growth in India. The well-being of the world can benefit by building infrastructure, political trust, and an economic boost in undeveloped countries, especially those in Africa. Furthermore, China has been known to have an abundance of coal, however, it is more economical to import methanol nowadays as opposed to producing it from coal. It is important for India to do this in order to imitate what China is doing. China is known to be one of the leading countries when it comes to methanol consumption and in the petrochemical industry. However, even China shifted away from the use of coal due to revolutionary technologies and is now focusing on importing methanol. Therefore, it is important for India to have their methanol imported in order to have their economy grow and therefore imitate what China is doing. Moving forward with socio-political trust, investing in countries like these will not only attract other investors from developed countries but will build relations in which the undeveloped countries will build a sustainable government. Instead of leaving countries poor and neglected, they need to be put on the map to help the local communities gradually become towns and eventually self-sustained cities.

Explanations of Goals and Objectives

The goal of this project is to focus on the social economical side of this process. The idea to make this process more efficient by focusing on the location of this plant and at the same time find a way to help end hunger and poverty by boosting poor economies in undeveloped countries. The objective in this project is to base the plant in India, which has the highest growing population in the world, and is in dire need of meeting its demand in order to avoid becoming dependent on other countries. To do this, a methanol plant will be invested and built in Angola, Africa. Doing this will help solve the previously stated problems in two major ways. First, investing a plant in Angola will help boost the economy and create new jobs. This could potentially be the initial steps into getting the country out of poverty. Second, the investment in the methanol plant in Angola will provide access to the methanol at a lower price than what India is currently paying the Middle East for Methanol. By doing this, MTO process will be economically optimized, due to a lower price of methanol, while also helping meet the increase in demand of ethylene and propylene in India. In conclusion, the goal of this project is to help meet the chemical demands of India due to its population growth, as well as assisting underdeveloped countries improve their economies by economically optimizing this process. A perfect way to solve many problems with one socially economic solution.

Market Analysis

In an MTO process, there is one raw material, which is methanol. Methanol can be produced primarily from coal or natural gas. In this project, the plan is to build a methanol plant in Angola, Africa and have methanol shipped over to the plant that will be built in India, specifically Mumbai. As of today, India, specifically West India, primarily obtains its methanol from the middle east and they have it shipped directly to on-site MTO plants. Methanol continues to grow globally in demand. The market size of methanol around the world, originally in 2018 was estimated at 1.81 billion USD and is projected to accelerate at a CAGR of 2.9% from 2019 to 2025.[4] Furthermore, the demand for methanol specifically for MTO process has risen significantly and will continue to rise in the upcoming years. MTO is one of the fastest growing processes for the end-use of methanol globally. In the plastic industry, ethylene and propylene are the olefins with the highest demand which is leading to plastic growth.[3] In *Figure 1*, it is shown how methanol needed for MTO is at very high demand as of today and continues to rise to the year 2025. Countries such as India while lie in the Asia Pacific region have been known to have a huge hold on the methanol market share.

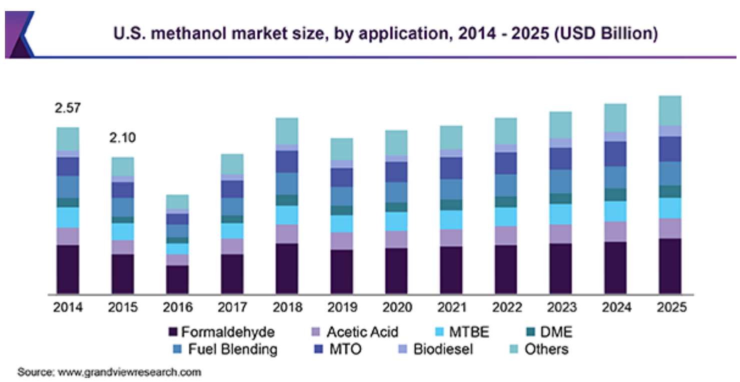

Figure 1: U.S. methanol market size [5]

As shown in *Figure 2*, Asia Pacific is the leading regional market in 2018, where it overtakes about 61.6% of the methanol revenue share.[5] Furthermore, methanol prices have been very low in Asia compared to different regions globally. This gives an idea of the large connection between methanol market and the Asia Pacific region.

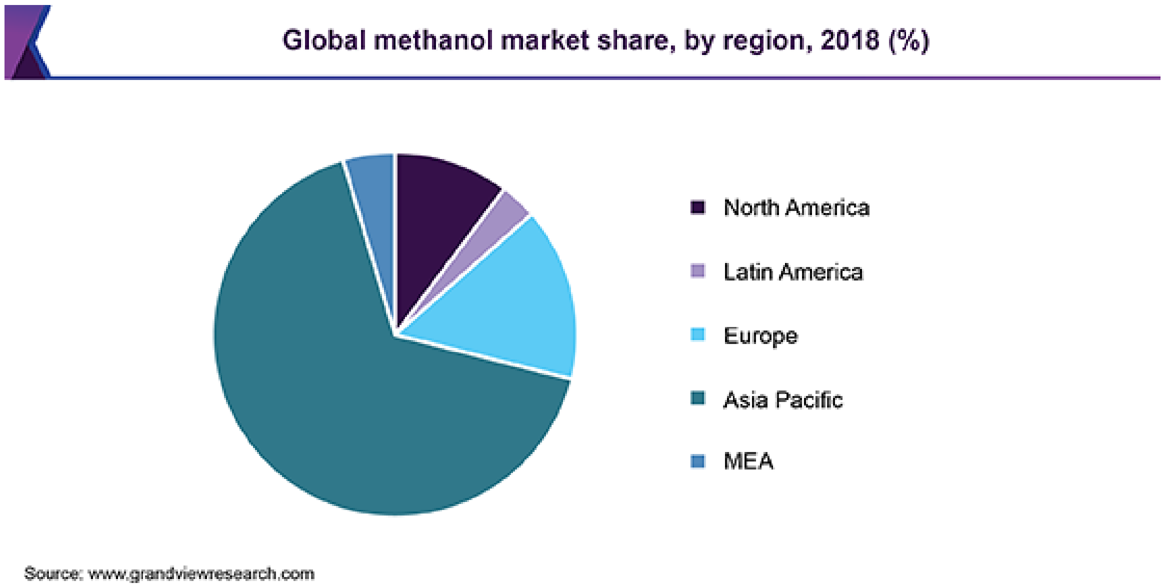

Figure 2: Global methanol market share.[5]

As shown in *Figure 3*, in the midpoint of 2019, methanol cost in Asia is approximately $275 per metric ton, which is more economical than using coal to make the methanol. Seeing this figure, there is a clear decrease in methanol prices which enhances the idea to be able to get methanol for an even cheaper prices from Africa to Asia.

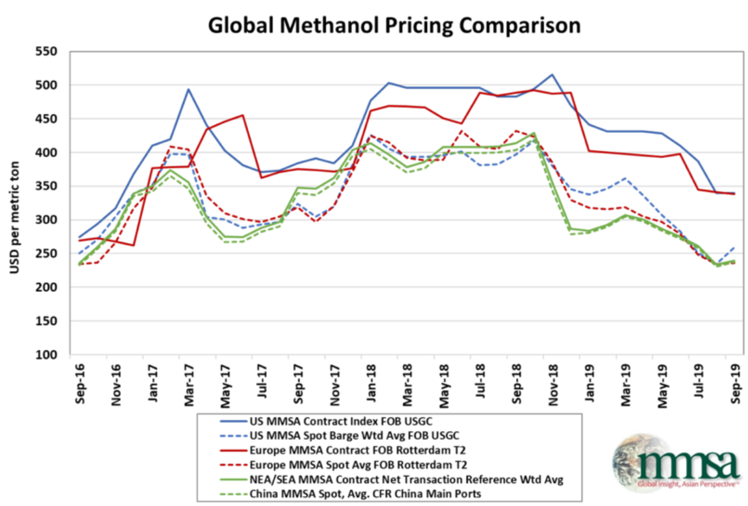

Figure 3: Global Methanol pricing Comparison[6]

This MTO process produces two main products, which are ethylene and propylene. Ethylene and Propylene are essential products that are need to for daily life. The largest volume in the petrochemical industry are olefins. As they are what is needed to make things such as packaging materials, piping and insulation, to synthetic rubbers and antifreeze.[1] This is essential in a country with a rapidly increasing population such as India. India, which is in the Asia Pacific region, has one of the largest capacities for olefins, specifically ethylene. As shown in *Figure 4*, the Asia Pacific region capacity for ethylene has risen significantly and looks to keep rising in the upcoming years versus other regions globally. Due to this, there is a visible decline in North America and Europe while other regions were growing in ethylene capacity from 1980 to 2020.

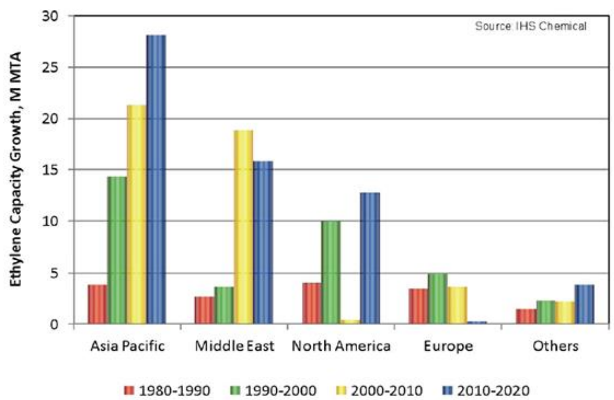

Figure 4: Global Ethylene Capacity.[1]

The market for these olefins in India is extremely large, with India producing between 9-10 million metric tons per year. They have such a high demand that in many forecasts the supply is not meeting the demand in the upcoming years, specifically in India to the large increase in population annually. As shown in *Figures 5* and *6*, the ethylene supply and demand are about to break even in 2023 while the propylene supply and demand is not enough to even meet the demand. This gives an idea of the increase demand in the olefins market, specifically in India. The demand for ethylene is supposed to grow at 11% CAGR and will reach 13 million metric tons by the year 2030.[3] Furthermore, as seen in the figures below by the year 2030 there will be approximately 23 million metric tons demanded of ethylene and propylene combined. This shows how much the demand in this market is growing and how a solution is essential in meeting these demands.

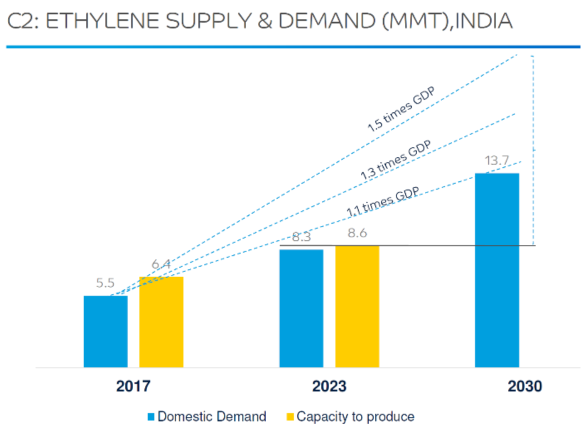

Figure 5: Ethylene Supply and Demand(India)[3]

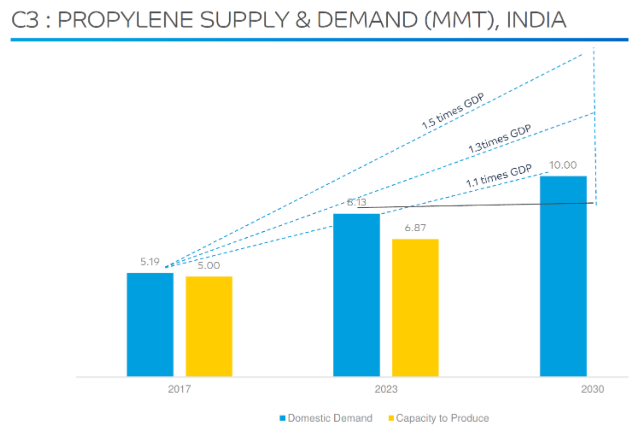

Figure 6: Propylene Supply and Demand (India)[3]

Environment Regulations:

One of the main regulatory issues that arose is the 2015 Paris agreement which states that Africa must decrease its carbon emissions by 2030. This was done because Africa is burning off its excess gas into the air causing heavy carbon emissions. This process and the location that was previously decided on will help Africa meet this agreement because the flue gas will be converted to methanol which will be used for the MTO process. More of the analysis will be put in detail in Safety and Regulations section.

Process Description:

What is the process?

The process that will be used is the Advanced MTO process. The Advanced MTO process is a combination of the UOP/Hydro MTO process and the Olefin Cracking Process. To fully understand the process, let's begin by looking at the chemistry that should occur to give us the greatest product yield. *Figure 7* shows the MTO mechanism and the different branches of olefins that it

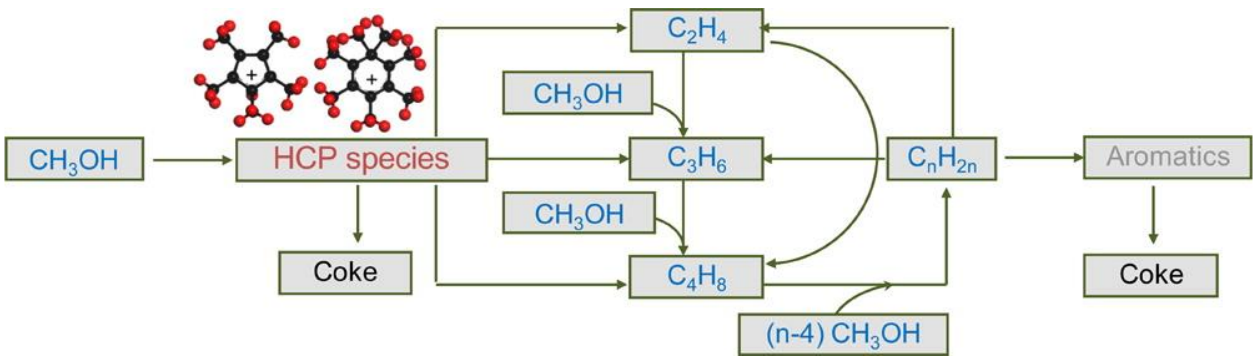

Figure 7: Methanol Reaction Mechanism [20]

can create. The feedstock methanol does not have to be pure and the catalyst will be the deciding factor of what selectivity and olefin yield will be produced. The hydrocarbon pool (HCP) mechanism consists of a zeolite cage channel that allows the assembly of olefins from methanol to be created.[20] These olefins include ethylene and propylene among other light olefins. With the different sized catalyst cations available, this allows us as the manufacturer to control coke formation, light olefins, or heavy hydrocarbons up to the desired specifications. Since the reaction is an exothermic process, there needs to be a system to control the temperature inside the reactor to prevent a runaway reaction. A runaway reaction will occur due to improper control of temperature which can cause an explosion in a piece of equipment. This will cause a significant financial loss to the economical side of the process. Since the catalyst is in application of volatile materials, the temperature of the reactor should be kept no higher than 550 °C.[20] The heat of reaction is around -196 kcal/kg at 495 °C so proper heat integration should be applied to remove the heat.[20] Due to possible coke formation and catalyst deactivation, a fluidized bed reactor will be superior over a fixed bed reactor as it allows the heat transfer to constantly be in motion to allow the dispersion of heat.

Chosen Catalyst

The choices of catalysts that can be used for the MTO process is SAPO-34 or ZSM-5. The difference between the catalysts is the size of the pores on the catalysts which allow either light or heavy hydrocarbons to diffuse. SAPO-34 contains a pore size of about 4 Å whereas ZSM-5 pore size is about 5.5 Å.[20] As shown in *Figure 8*, the small pore sizes of SAPO-34 will not allow the big hydrocarbons that are produced to diffuse in any way. Doing this will lead to the smaller olefins such as ethylene and propylene to have a higher selectivity.[13]

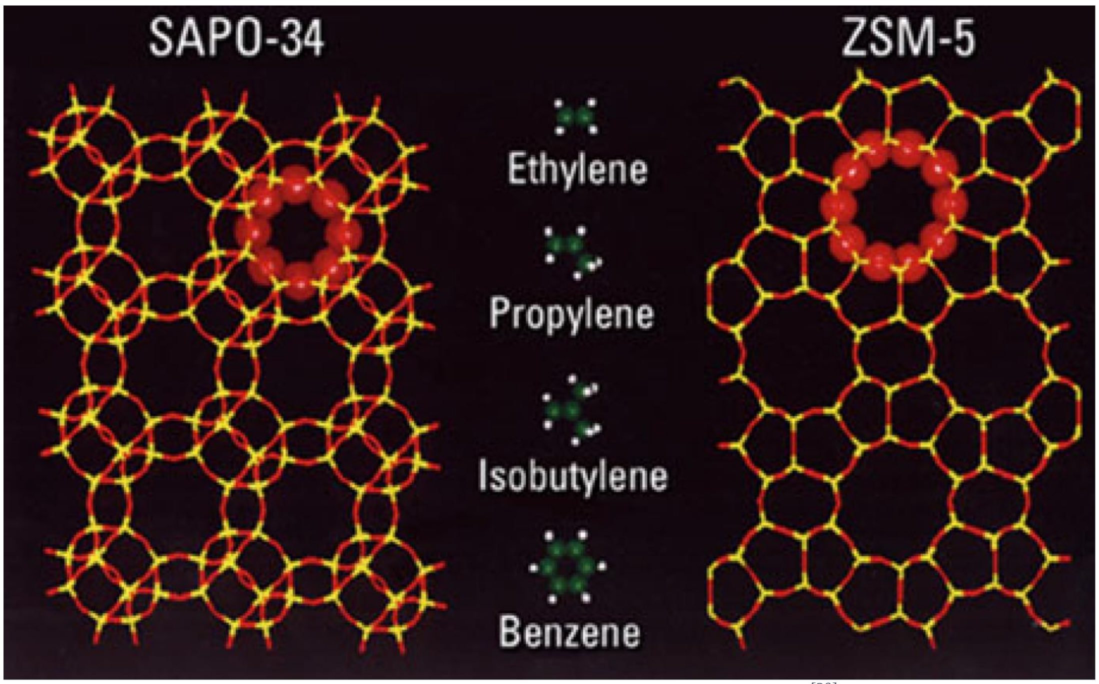

Figure 8: SAPO-34 and ZSM-5 Catalyst Comparison [20]

ZSM-5 will not be a viable choice as it has large pore openings, which leads to the large hydrocarbons to diffuse and a smaller selectivity of the light olefins which are the ones that are desired.[6] In this process the best catalyst that can be regenerated and at the same time will lead to the best conversion of methanol to the olefins products is the SAPO-34 catalyst.[13] The regeneration factor of the catalyst is important as this will change the overall operational costs that will be incurred when running the reaction. Another feature that shows the strength of this catalyst is the optimized acidity it has. This simply decreases the amount of paraffinic (branched-chained hydrocarbons) by-products, which is evidently the reason why the advanced MTO process has a 75-80% selectivity of ethylene and propylene on a carbon basis without any splitter columns.[14][20]

Chosen Reactor

Considering the type of catalyst, the exothermic temperature of the reactor, and the heat integration needed, the clear choice of reactor for this process is a fluidized bed reactor. When considering a fixed bed reactor, the removal of the catalyst will be required more and the cost of utilities to cool down the reactor will be larger than a fluidized bed. In addition, the fixed bed reactor will need two reactors to keep the process going; one reactor where the reaction takes place, and the other reactor moves the deactivated catalysts that need to be transferred to a catalyst regeneration reactor. A total of 3 reactors will be needed for the fixed bed reactor.

The fluidized bed reactor will only need a total of 2 reactors. One reactor will be where the mechanism will occur and the other for regeneration of the catalysts. Rather than having an extra reactor vessel solely for the deactivated catalysts, the fluidized bed reactor will allow for the control of reaching a steady flow of deactivated catalysts and regenerated catalysts to circle around the system. Since the reactor is in fluidized state, heat will be dispersed throughout the reactor and will reduce the costs of utilities compared to a fixed bed reactor. Furthermore, since a fixed bed reactor would have trouble maintaining this constant heat transfer motion, this might lead to purchasing another reactor in order to maintain this flow. Therefore, the fluidized bed reactor is also more cost efficient in this way.

Process Explained with Operating Conditions

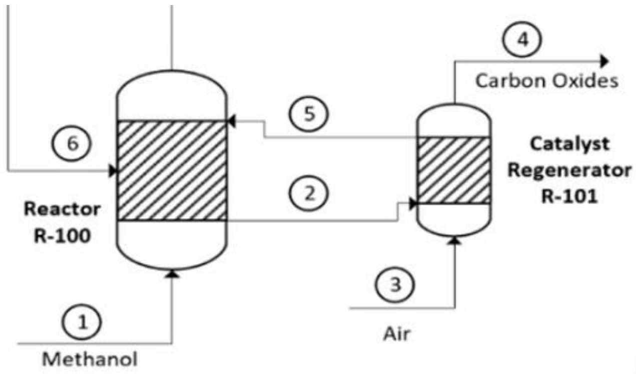

Figure 9: Reactor Section PFD

The process begins with methanol feed (stream 1) entering the fluidized bed reactor shown in *Figure 9*. The reaction that takes place inside the reactor with the catalyst is:[20]

$$2CH_3OH \rightarrow CH_3OCH_3 + H_2O \tag{1}$$

Reaction 1 shows how the methanol forms DME and water but DME reacts immediately again to form ethylene (reaction 2), propylene (reaction 3), and C_4^+ heavy hydrocarbons (reaction 4):[20]

$$CH_3OCH_3 \rightarrow C_2H_4 + H_2O \tag{2}$$

$$3CH_3OCH_3 \rightarrow 2C_3H_6 + 3H_2O \tag{3}$$

$$4CH_3OCH_3 \rightarrow 2C_4H_8 + 4H_2O \tag{4}$$

With the SAPO-34 catalyst, the heavy hydrocarbons are in trace amounts and will not diffuse in great amounts. The fluidized bed reactor will operate at around 450 °C and roughly 2 bar.[14] This low pressure that methanol is entering the reactor will lead to a high selectivity of the desired product which is the light olefins. Furthermore, a high temperature is needed because it will evidently lead to a higher yield of ethylene.[13] The catalyst in the reactor will deactivate over time of the process, so a good portion of it is sent to a catalyst-regeneration reactor unit (R-101) via stream 2. This reactor which also has an air feed will regenerate that portion of the deactivated catalyst and send it back into the fluidized bed reactor in stream 5.

This will lead to a product stream which contains water and DME (stream 7) sent to a separator unit which removes the water (stream 8) shown in *Figure 10*. The DME and the impurities are sent to separation units where the DME is recovered (stream 6 shown in *Figure 9*) back to the reactor. The resulting product is sent to stream 10 to a mixer to be separated.

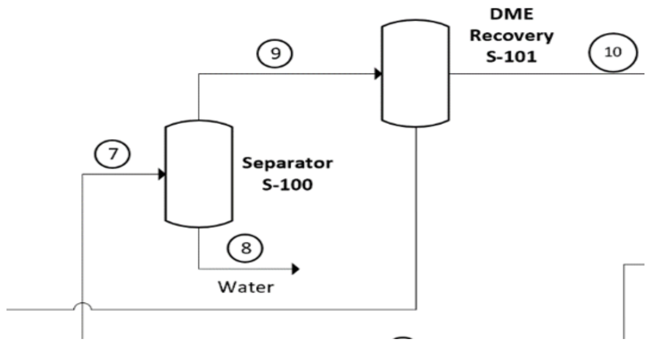

Figure 10 – Separation and DME Recovery Section PFD

Figure 11 shows the product stream being mixed into stream 12 that goes into the separation process. This is shown as a single unit called fractionation but consists of about 4 separation units in order to separate ethylene, propylene, propane, ethane, fuel gas, and other C_4 products.

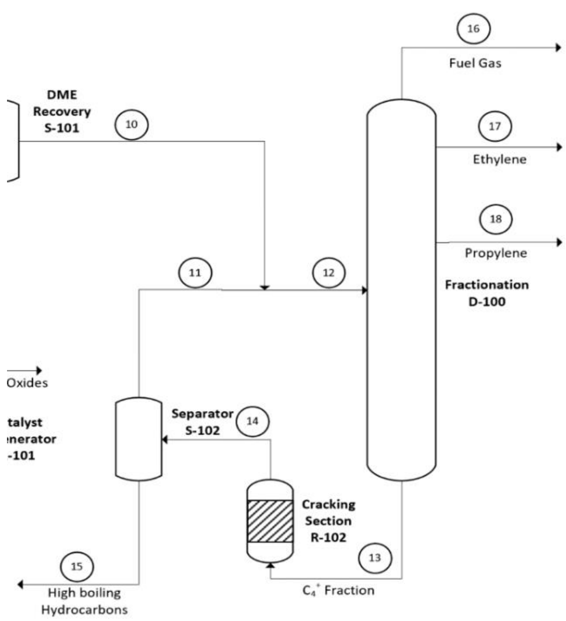

Figure 11 – Separation and Cracking Section PFD

The fractionation unit will be operating between 10-11 bar and with low temperature of about 10-20 °C. Stream 13 shows the C_4 heavy hydrocarbons that will go into a cracking section to recycle them back into the fractionation process. The cracking section will separate light olefins such as methane (reaction 5), ethane (reaction 6) and propane (reaction 7):[20]

$$CH_3OH + H_2 \rightarrow CH_4 + H_2O \quad (5)$$

$$C_2H_4 + H_2 \rightarrow C_2H_6 \quad (6)$$

$$C_2H_4 + H_2 \rightarrow C_2H_6 \quad (7)$$

The hydrogen gas comes from the carbon monoxide reacting with water molecules in the air for the light olefin cracking. Stream 14 will separate those light olefins from the heavy olefins that will lead to stream 15 getting rid of the big hydrocarbon chains. This will ensure high selectivity of desired products are separated into fractionation. This process is essential in making sure the desired ethylene to propylene ratio and selectivity is met. The ratio between ethylene and propylene when produced at the highest yields is between 1.75-2.[13] The selectivity of methanol will be close to 90% for this process.

What are possible solutions/Competing Processes?

There are many processes that produce olefins and can have the same outcome as MTO. Those different process include Steam Cracking, Oxidative Coupling of Methane, Fisher-Tropsch Synthesis and Catalytic Dehydrogenation. One of the previously stated process that will be discussed is Steam Cracking, which is an exothermic process that has no catalyst. Steam Cracking is a widely used technology which is heavily dependent on naphtha cracking, and naphtha cracking is dependent on crude oil. Steam Cracking can take a saturated hydrocarbon and break it down into smaller, unsaturated, hydrocarbons.[15] This entails taking a single bonded carbon chain molecule and turning it into a smaller carbon chain with at least one double or triple bond. This process works by taking a feed stream consisting of a gaseous or liquid hydrocarbon, ethane or liquid pressurized gas as examples, and dilute the stream with steam, that is then heated in a furnace, at approximately 850 °C, for a few milliseconds.[15] Having those conditions for the reaction help improve the yield of the wanted product. An advantage to Steam Cracking, as stated before, is the method is well-established and widely used. Another is that the ratio of ethylene and propylene can be manipulated, but only by using two different crackers. Since the price of coal and natural gas are cheaper and more economically feasible, it would make using either ethane or naphtha cracking a disadvantage. Also using two different cracking units in order to manipulate the ratio of ethylene to propylene would be too costly making it a disadvantage to use. In the generalized chemistry below, it shows the two steps it takes to make the ethylene product. The first step is taking propane and breaking it down into ethane and methane. The second step is the cracking step which takes the ethane and turns it into ethylene.

$$RCH_2CH_2CH_2R' \rightarrow RCH_2CH_2 + R'CH_2 \tag{4}$$

$$RCH_2CH_2 \rightarrow R + CH_2 = CH_2 \tag{5}$$

Another possible process is Oxidative Coupling of Methane (OCM). This method would be a good option if you wanted higher hydrocarbons.[16] Despite the method giving higher hydrocarbons it is not favored because there was never any success in the method causing many to only dream of having this process work. There were many limitations, but regardless of those drawbacks it would be a great technological advance. To this day there is still research going on, but to no avail. As previously stated, there are many flaws to the process and some include a low conversion rate, slow catalyst regeneration, a high temperature range of about 750 – 950 ˚C, and a lack of skill to ensure the process goes on without any problems.[16] An advantage for the OCM process is that the catalyst at higher temperatures give higher stability. A couple disadvantages to the OCM process include giving a low yield of ethylene and is highly exothermic. The chemistry shows that two molecules of methane is converted into ethane and a hydrogen molecule, while the second equation shows the same process but with water in the product by having the methane react with an oxygen molecule.

$$2CH_4 \rightarrow C_2H_6 + H_2 \; \Delta G^{\circ}(1000K) = 71\frac{kJ}{mol} \tag{6}$$

$$2CH_4 + \frac{1}{2}O_2 \rightarrow C_2H_6 + H_2O \; \Delta G^{\circ}(1000K) = -159.7\frac{kJ}{mol} \tag{7}$$

The third process is Fischer-Tropsch Synthesis and is a better process for making gases into liquid or GTL. The yield of propylene is only about 10% carbon fraction and produces more gases than liquid. The Fischer-Tropsch Synthesis is a catalyzed chemical reaction with syngas being made into substances like ethane, liquid pressurized gases, and naphtha hydrocarbons.[17] Some hydrocarbons are in the gaseous state at room temperature, while others are in the liquid or solid state, also hydrocarbons are usually nonpolar and not dissolvable in water. Fischer-Tropsch Synthesis is desired for its use of various environmental materials like biomass, coal, and natural gas to make chemicals and fuels.[17] This would prolong the sustainability of the demand for liquid fuels and make less use of the dwindling petroleum reserves. The advantages to the Fischer-Tropsch Synthesis process include, but are not limited to, offering a diverse market to natural gas resource holders and convert the syngas into ultraclean fuels. Ultraclean fuels are fuels that lack the presence of aromatics, sulfur, or nitrogen compounds.[17] The disadvantages of the FTS process are it's too expensive to be profitable and there is a low selectivity. The generalized chemistry of this process is shown below, with the reactions for paraffins, olefins, and alcohols, respectively.[18]

$$\boldsymbol{nCO + (2n+1)H_2 \rightarrow CnH_2n + 2 + nH_2O} \tag{8}$$

$$\boldsymbol{nCO + 2nH_2 \rightarrow CnH_2n + nH2O} \tag{9}$$

$$\boldsymbol{nCO + 2nH_2 \rightarrow CnH_2n + OH + (n-1)H_2O} \tag{10}$$

The final competing process is Catalytic Dehydrogenation, which has recently come into play since around the year 2012. Prior to the year 2010 there was no propane dehydrogenation, but that all changed in 2012 with the finding of shale gas. Catalytic Dehydrogenation is now widely used especially for propane because of the abundance of it to make propylene from the propane dehydrogenation unit and the same can be said for isobutylene from the isobutane dehydrogenation unit. Catalytic Dehydrogenation is a process in which produces a higher amount of a single olefin due to a higher selectivity for that olefin, like propane and isobutane. Some great advantages to using the Catalytic Dehydrogenation process is that there is an abundance of cheap light olefins produced from shale gas, and a high profitability due to the price of propane being lower compared to propylene.[19] A disadvantage is that it is more selectivity of propylene. The chemistry of this process is shown below.

$$\mathbf{C_2H_6 \leftrightarrow C_2H_4 + H_2} \tag{11}$$

$$\mathbf{C_2H_6 + 12O_2 \leftrightarrow C_2H_4 + H_2O} \tag{12}$$

$$\mathbf{C_3H_8 + 2O_2 \leftrightarrow CH_3 = CH - COOH + 2H_2O} \tag{13}$$

What is your solution?

The chosen solution is the UOP Advanced MTO Process. The process has a higher yield than the other existing methanol technologies. The average MTO plant worldwide has a carbon yield of about 75% while the Advanced MTO plant has an average carbon yield of 90%.[1] This means the process is more environmentally friendly than the competing methanol technologies available.

Why you have chosen that solution?

Since the import of methanol will be coming from the region around Angola, the cost associated with shipment of raw material needs to be considered. The process will not require the methanol to be pure as the plant can run with un-distilled methanol.[1] This will lower the capital cost of the Angola refinery of natural gas to methanol. With a high carbon yield being produced in the Advanced MTO process in India, this will clean any of the impurities and will remain at a higher yield than competing processes. The goal of the project is to establish a robust relationship between India and Africa. The socio-economic implications will lead to a sustainable economy that will bring in jobs, new infrastructure, hospitals, schools, and stronger communities to the people in African countries. Flue gas in Africa is simply burned off as there is no use for it. Once India establishes a contract for the purchase of methanol, Africa will be able to use the flue gas to convert it to methanol. This will decrease the carbon footprint in Africa as they are under contract for the 2015 Paris Agreement to lower emissions by 2030.[7] To avoid hefty fines, it will be in both Angola's and India's best interest to establish this project. One will benefit from substantially improving their economy and the other from the purchase of low-cost methanol to produce the needed olefins. If the worst-case scenario of having un-distilled methanol was examined, the MTO plant will be able to convert it to high yield. The benefit of having this technology is the zero-emission rating the plant can produce. The values revolve around the 90% yield with 10% yield being the water produced in the process. This water will then be stripped of heavy carbons and used as irrigation system for the farms in India. The heavy carbons will then be recycled back to the catalyst reactor to regenerate the process and avoid coking of materials.

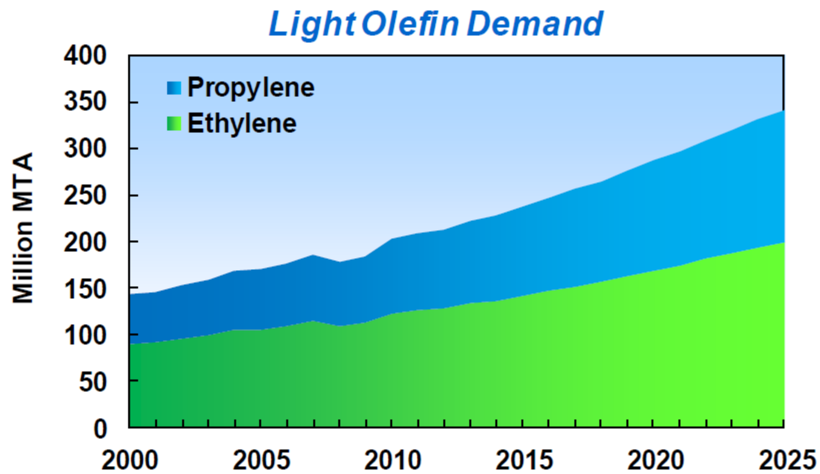

Figure 12: Light Olefin Demand[9]

Since the demand in India for propylene and ethylene is increasing greater than the global demand, it is beneficial to meet and exceed the demand in India to meet the trend *Figure 12*.[9] At 8-10% growth, the growing population of India will meet the requirements to sell the product. Selling ethylene or olefins to surrounding countries in India can also be a viable solution as the Advanced MTO process has by-products that can be sold. Another reason that this process was chosen is due to the ability to upscale the production of building more plants or increasing the reactor diameter to increase capacity. This will reduce the capital cost and make India's plan of building more MTO processes lucrative. The scalability of the project will be to meet the demands of about 600,000 tons/year of new India demand of ethylene and propylene. This will be met with the Advanced MTO process.

Process Control

Ethylene and Propylene Production

Methanol to Olefin Main Reactor (R-101):

The main reactor is extremely exothermic due to the conversion of methanol to olefins. Run-away reactions need to be taken seriously as it can cause serious injury to people and damage to the surroundings. To prevent run-away reactions, the reactor needs to be considered as isothermal at 450 °C by manipulating the amount of catalyst regenerated and using a catalyst cooler system. Additionally, high temperatures will affect the conversion rate of methanol to SAPO-34 which will affect the purity of the end products.

Methanol will be introduced into the reactor at a temperature lower than 350 °C due to metallurgical properties according to Mr. Vora. The methanol temperature will be controlled in the vaporizer which will be a heat exchanger from the reactor effluent stream that will heat up the methanol feed. The catalyst will also be operating at isothermal conditions between 350-400 °C by taking portions of the catalyst continuously to the catalyst cooler and then putting it back into the reactor. The cooled catalyst will then be reintroduced into the reactor vessel and will essentially cool down the other catalyst present to remove the heat. Cooling duty will be calculated by getting close to the change in temperature from the methanol feed and the reactor temperature of 450 °C. The heat of reaction of methanol will have the vaporizer with water going in at about 50 °C and bringing it up as steam at 400 °C. This steam can then be used to power either the compressors, create electricity, or sell to nearby plants.

There is a slight pressure drop in the reactor of about 0.5 bar to create cyclones inside the reactor. The purpose of the cyclones is to minimize the dust going to the overhead of the reactor. This is achieved by slightly increasing the diameter at the upper portion of the reactor and the linear velocity drops substantially at the top to drop the dust to the bottom. This allows for dust particles of about 5 microns or less to go to the top and bringing anything bigger to the bottom. This helps the quench tower from having a lot of catalyst particles. The pressure inside the reactor is not controlled at the inlet but rather by compressor suction at the top so for any pressure changes, this can be a factor of changing the inner pressure from 1.9 bar to 2.1 bar with an operating pressure of 2 bar. A release valve will be attached to vessel to prevent over pressurization of the reactor.

WHSV controls the conversion of methanol from SAPO-34. Achieving a conversion of around 98-99.5% at 1.5 hr^{-1} is ideal as having less conversion rate will need to oversize most of the equipment in the MTO process and will lead to less efficiencies throughout the process.[7] Therefore, a control of 1.4-1.6 hr^{-1} is needed to get the ideal conversion rate.

The benefits of having control in the temperature through the catalyst cooling is that changes in selectivity can be achieved. By changing the temperature between 500-520 °C higher production of ethylene can be achieved whilst reducing the ratio of propylene if needed. Lowering the temperature to 400-430 °C increases the selectivity of propylene with less ethylene.[7] Having to change only the temperature without changing the whole process is ideal if market swings allows for one of the chemicals is more profitable.

Another factor to consider in the main reactor is coking on the catalyst. This is controlled by manipulating the regeneration flow rate. If conversion is lower at the main product end, assuming all other equipment is working as it should then there could be a problem with catalyst having too much coke. Another factor that can be identified is the reactor is giving off too much heat. This can give an indication that the regenerative flowrate will need to be increased.

Although the catalyst regenerator was not simulated, it will be similar to that of the main reactor where it will be isothermally operating from 550-560 °C and the temperature will be maintained by independent cooling system of introducing new catalyst from main reactor. The coke is assumed to be at steady state somewhere in the region of 5% coke and will come out of the regenerator at 0% and again, cooled down to 450 °C using cooling water. However, the catalyst regenerator plays a role in keeping the main reactor in isothermal conditions since it will be burning off the coked catalyst. The catalyst regenerator will also need a release valve for any pressure changes.

Table 3: R-101 Control Variables

Control Variable	Set Point	Lower Limit	Upper Limit	Manipulated Variable
Temperature (°C)	450	425	500	Catalyst Coolant
Pressure (bar)	2	1.9	2.1	Product Flowrate
WHSV (hr^{-1})	1.5	1.4	1.6	Feed Flowrate

Quenching Process (S-300 & S-301):

The main purpose of the quenching process is to get rid of the impurities and the large amount of water produced. The amount of water produced by weight of methanol is a little under 57% which is why the system is needed to be split up into two different sections. The first one will essentially filter out particles between 0 and 10 microns into what is called a 'cake.' This cake will be filtered out by landfill from an outside company that is out of scope from this project. Since water is soluble with methanol, this will need to be distilled to strip away hydrocarbons, DME, methanol, and other impurities from water. Once the water goes to the water effluent management or a water treating tower for purification, it will be clean enough to be used for irrigation in India's farms. This can essentially be sold to farmers as irrigation quality water. Additionally, some of the water can be used within the process to cool down reboilers.

To remove the impurities, a neutralizing agent can be introduced at the top of the tower to get rid of these hydrocarbons, particles, and acid compounds such as $Ca(OH)_2$ or NaOH. The ratio that is preferably used of neutralizing agent to water is 0.38:1.[7] Therefore, a control of 0.37:1 and 0.39:1 will need to be put in place to have control. This will allow for the purification process to be minimized as it will not need unnecessary equipment down the line.

The temperature out of the first quench tower should have an optimized temperature according to literature of around 35 °C. The upper and lower bounds range from 45 to 25 °C respectively. Alarms will be placed in the unit to indicate any fluctuations outside of the range for them to be corrected by coolant flowrate. Due to pressure loss of vessel, this will be operating at around 1.8 bar ideally and again will both need safety release valves.

The next quench tower S-301 will function as a dryer to get rid of all the water in the product stream. The first tower essentially separates the impurities from the gas. The second tower will remove all the water by partially condensing the reactor effluent which will make the top the light olefin vapor and the bottom will be pure aqueous solution that will be purified further to irrigation quality. Before being purified, the aqueous solution will go through a stripper to be put back in the first and second quench towers at low quantities as some recycled water is needed. This will be controlled depending on the coking collected in the first filtration step.

Table 4: S-300 & S-301 Control Variables

Control Variable	Set Point	Lower Limit	Upper Limit	Manipulated Variable
Temperature (°C)	35	25	45	Coolant Flowrate
Pressure (bar)	1.8	1.7	1.9	Pressure Release Valve
Composition Ratio (neutralizing agent:water)	0.38:1	0.37:1	0.39:1	Neutralizing agent Flowrate

<u>C2 Splitter (S-401):</u>

The C2 Splitter (S-401) will have three different kinds of control system loops that will be talked about for the condenser and reboiler. The three different control system loops are: temperature, pressure and flow rate.

In the condenser, portion of S-401 pressure will be monitored/controlled. The first instance is the ethylene exit stream, which consists of a pressure valve attached to the stream and a pressure indicator controller and a pressure transmitter connected to the top stream going to the heat exchanger, H-400. The pressure will drop 2 bars going from 20 bar to 18 bar and this is achieved by the assistance of the pump. The second instance is the overhead pressure release, which will consist of a pressure valve with a pressure indicator controller and pressure transmitter connected to S-401 itself. Next temperature will be controlled through H-400. The coolant, propylene, flows through H-400 consisting of a temperature control loop with a temperature valve attached to the coolant stream with a temperature indicator controller, temperature transmitter and a temperature element connected to the stream going to the shell and tube heat exchanger, A-400. The pressure control loop in the first instance connected to H-400 influences temperature. The affect being a drop in temperature going from -30 ˚C, the temperature of the stream going into S-401, to -33 ˚C of the exit stream of ethylene. Lastly, the flow rate is controlled and monitored with level alarms. The flow rate contains a control loop consisting of a flow valve attached to the stream going back into the top of S-401, with a flow indicator controller and a flow transmitter connected to the stream right before the valve, and a level indicator controller connected to the flow indicator controller and A-400. The level indicator controller has a high- and low-level alarm for any mishaps that could happen. Some mishaps include any faulty or leaking pipes or in the adverse a clogged pipe, and some precautions that could be considered to solve these issues include routine maintenance checks and removing the clogs.

In the reboiler, portion of S-401 temperature will be monitored/controlled. Temperature is monitored with the use of high- and low-level alarms and is controlled with a temperature control loop. It consists of a temperature valve attached to the water stream going into the heat exchanger(H-401) with a temperature indicator controller, temperature transmitter and a temperature element connected to the bottoms of S-401. A high- and low-level alarm is attached to the temperature controller because there could be serious adverse effects of either a runaway reaction or a lower purity of the exiting ethane stream, which could skew the purity of the product. A possible remedy to these is a routine maintenance check. The other aspect to be monitored/controlled in the reboiler is the flow rate, and that is done by having a flow valve attached to the exiting ethane stream with a flow indicator controller and flow transmitter connected back to the bottoms of S-401, with a level indicator controller connected to the flow indicator controller. The level indicator controller has a high- and low-level alarm to warn if there could be a possible leak in the pipes or a clog, which could cause a problem in S-401. Possible remedies to these problems are regular maintenance checks and removal of the clogs.

Table 5: S-401 Control Variables

Control Variable	Set Point	Lower Limit	Upper Limit	Manipulated Variable
Pressure (bar)	20	18	22	C2 Flowrate
Temperature (°C)	-30	-33	-27	Coolant Flowrate

Depropanizer (S-402):

The Depropanizer (S-402) will have three different kinds of control system loops that will be talked about for the condenser and reboiler. The three different control system loops are: temperature, pressure and flow rate.

In the condenser portion of S-402 pressure will be monitored/controlled, and the first instance is the propylene exit stream. It consists of a pressure valve attached to the stream and a pressure indicator controller and a pressure transmitter connected to the top stream going to the heat exchanger (H-402). The pressure will drop 2 bars going from 20 bar to 18 bar and this is achieved by the assistance of the pump. The second instance is the overhead pressure release, which will consist of a pressure valve with a pressure indicator controller and pressure transmitter connected to S-402 itself. Next temperature will be controlled through H-402 with the use of a coolant, ethylene. It flows through H-402 consisting of a temperature control loop with a temperature valve attached to the coolant stream with a temperature indicator controller, temperature transmitter and a temperature element connected to the stream going to the shell and tube heat exchanger (A-401). The pressure control loop in the first instance connected to H-402 influences the exiting temperature. The affect being a drop in temperature going from 21 ˚C, the temperature of the stream going into S-402, to 14 ˚C of the exit stream, propylene. Finally, the flow rate is controlled and monitored with level alarms. The flow rate contains a control loop consisting of a flow valve attached to the stream going back into the top of S-402, with a flow indicator controller and a flow transmitter connected to the stream right before the valve, and a level indicator controller connected to the flow indicator controller and A-401. The level indicator controller has a high- and low-level alarm for any possible problems that could arise. Some problems may include leaking pipes or in contrary a clogged pipe, and some precautions that could be considered to solve these problems include routine maintenance checks and removing the clogs.

In the reboiler portion of S-402 temperature will be monitored/controlled, and it is monitored with the use of high- and low-level alarms. These alarms are controlled with a temperature control loop consisting of a temperature valve attached to the water stream going into the heat exchanger (H-403) with a temperature indicator controller, temperature transmitter and temperature element connected to the bottoms of S-402. The alarms are attached to the temperature controller because there could be serious adverse effects of either a runaway reaction or having a bad separation of propylene from the heavier hydrocarbons. A possible remedy to these is a routine maintenance check. The other aspect to be monitored/controlled in the reboiler is the flow rate. This can be done by having a flow valve attached to the exiting 'HEAVY' stream with a flow indicator controller and flow transmitter connected back to the bottoms of S-402, with a level indicator controller connected to the flow indicator controller. The level indicator controller has a high- and low-level alarm to alert of any possible leaks in the pipes or clogs, which could cause problems in S-402. Some possible solutions are regular maintenance checks and removal of the clogs.

Table 6: S-402 Control Variables

Control Variable	Set Point	Lower Limit	Upper Limit	Manipulated Variable
Pressure (bar)	20	18	22	C3 Flowrate
Temperature (°C)	21	19	23	Coolant Flowrate

<u>C_2/C_3 Splitter (S-400)</u>:

The C2/C3 Splitter (S-400) will have three different kinds of control system loops that will be talked about for the condenser and reboiler. The three different types of control system loops include: temperature, pressure and flow rate.

In the condenser, portion of S-400 pressure will be maintained at a constant rate and that will be achieved with a pressure control loop. The loop will consist of a pressure valve attached to the C2 exit stream, going to S-401, with a pressure indicator controller and a pressure transmitter connected to the top exit stream of S-400 going to the heat exchanger (H-404). Another area that has a pressure control loop is the pressure release stream, having a pressure valve with a pressure indicator controller and pressure transmitter connected to the top of S-400. Next, temperature will be discussed as having a coolant flowing through H-404 to cool the stream down to -30 ˚C, going to S-401, from 92 ˚C coming from the stream MIX-C. The dramatic change in temperature is needed to get the correct separation of C2 from C3. This is possible by the control loop consisting of a temperature valve, temperature indicator controller, temperature transmitter and temperature element all connected to the stream coming out of H-404 and going to the shell and tube heat exchanger, A-402. The flow rate is controlled by a flow control loop consisting of a flow valve attached to the MIX-C stream, with a flow indicator controller and flow transmitter connected to the condenser, while having a level indicator controller attached to the flow indicator controller with a high- and low- level alarm for any possible leaks or clogs. These could be solved with a routine maintenance check and removal of the clog.

In the reboiler portion of S-400 temperature will be controlled by having cooling water flow through the heat exchanger (H-405) to cool the stream down to 21 ˚C, going to S-402, from 92 ˚C coming from the stream MIX-C. Again, the dramatic change in temperature is needed to get the correct separation of C3 from C2. This is done by having a temperature valve, temperature indicator controller, temperature transmitter and temperature element all connected to S-400, with high- and low- level alarms to monitor any possible runaway reactions or a bad separation of C3 from C2. These could be maintained by having regular maintenance checks. Flow rate is controlled by having a flow valve attached to the exiting stream of C3, with a flow indicator controller and flow transmitter connected to the bottoms of S-400. Connect to this is a level indicator controller with a high- and low-level alarm to alert to any leaking pipes or possible clogs. Again, these could be solved with routine maintenance checks and removal of the clog.

Table 7: S-400 Control Variables

Control Variable	Set Point	Lower Limit	Upper Limit	Manipulated Variable
Pressure (bar)	20	18	22	C2/C3 Flowrate
Temperature (°C)	92	83	101	Coolant Flowrate

Process Control Loop Schematic

C2 Splitter Unit – Before HAZOP

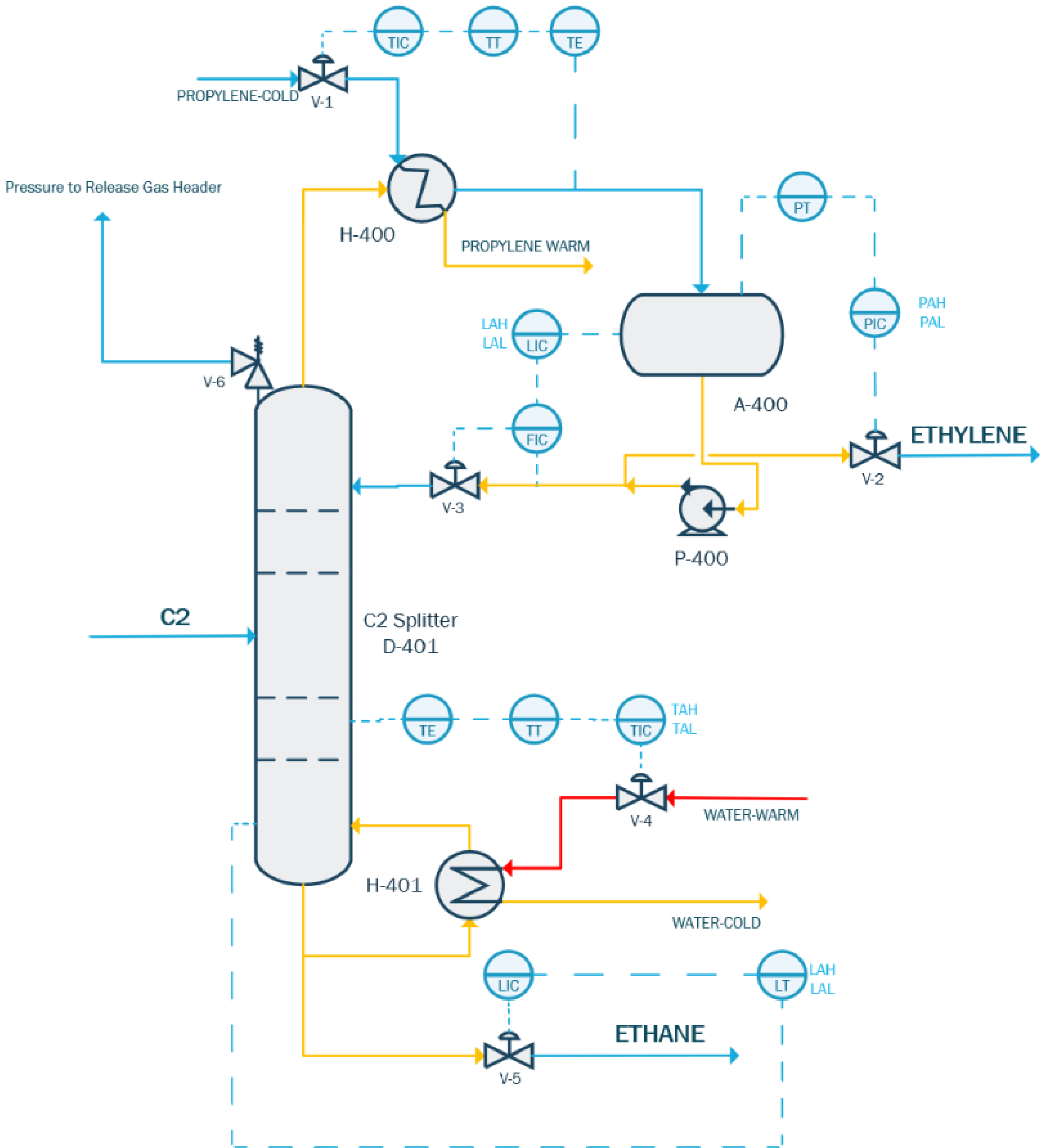

Figure 13: C2 Splitter Overview before HAZOP

The bottom section of the C2 splitter will consist of controls for the reboiler temperature and level control. This is to prevent column from flooding through valve V-5 with a level indicator controller and level transmitter. This will have a high- and low- level alarms that will control the inside aqueous levels. Temperature in the reboiler is important since this will separate the ethane from the ethylene being produced. To control this, valve V-4 will be controlled by the temperature indicator controller to a temperature transmitter and a temperature element inside the column. Controlling the bottom temperature is important in order to maintain proper product composition of ethylene and to minimize the contamination of ethane to the top of the column. Again, temperature will have a high- and low- level alarms in order to control the temperature remotely.

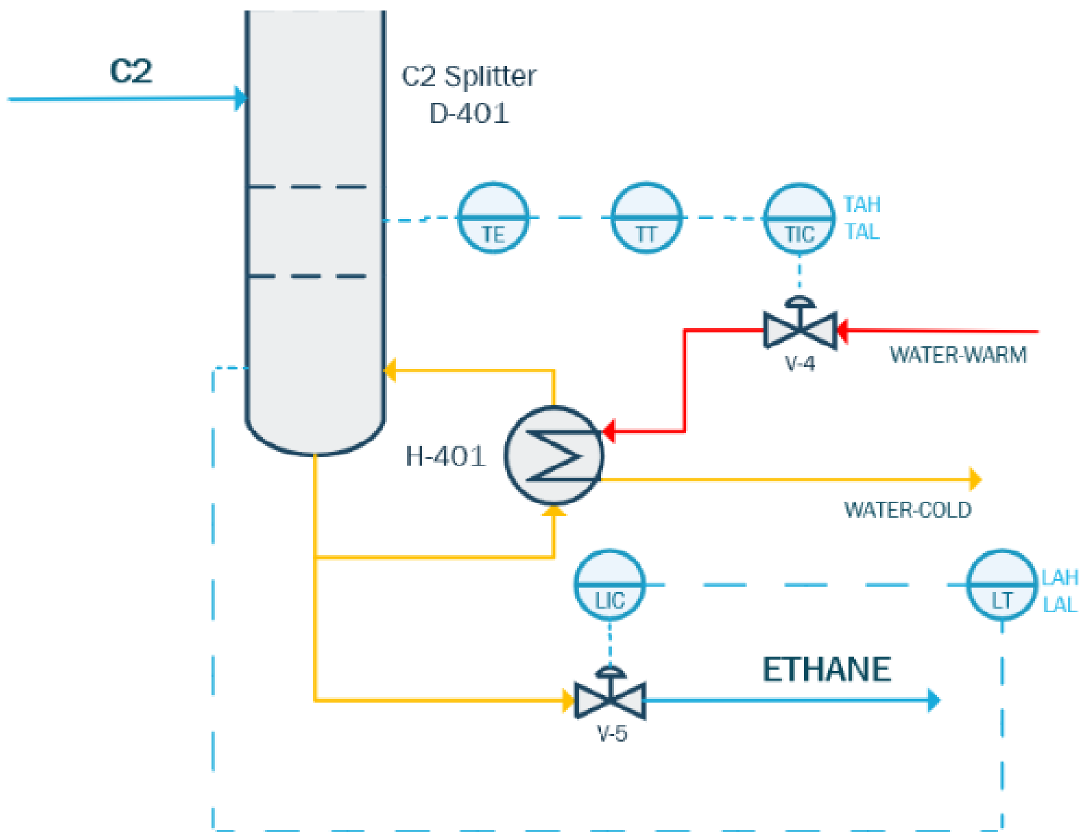

Figure 14: Bottoms Before HAZOP

The top of the column will consist of a spring valve V-6 normally placed on top of the column to automatically release pressure in the case it goes over the design specification. This is safely released to the gas header where it is burned off (out of scope). The temperature of the distillate is important to be controlled as this will give us proper vapor/liquid mixture in the accumulator A-400. This is controlled through valve V-1 using temperature control system. A-400 will have two different control systems, one will be the ethylene product control through pressure control with high- and low- level alarms in order to provide the correct composition at 19-20 bar. The reboiler ratio will also be controlled through the accumulator in valve V-3 by ensuring proper level is controlled. This will depend on the composition projected percentage of 99.4% through the atomization of ethylene in vapor and liquid form.

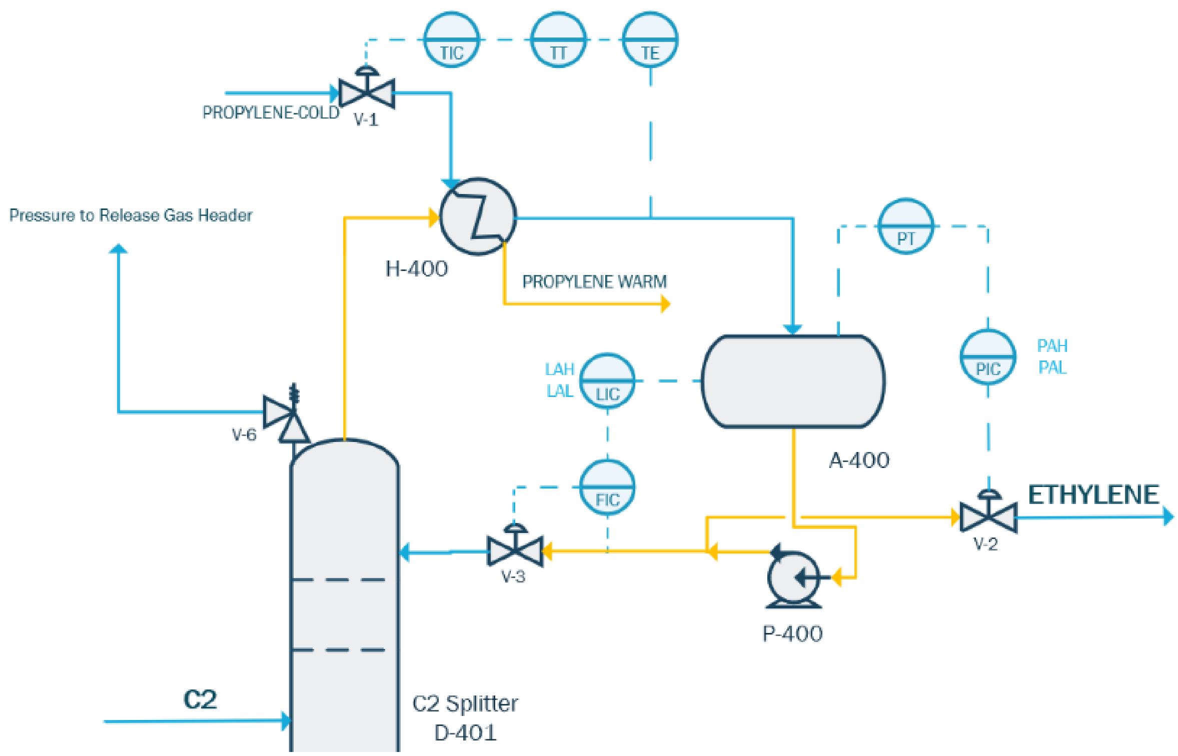

Figure 15: Tops Before HAZOP

C2 Splitter Unit – After HAZOP

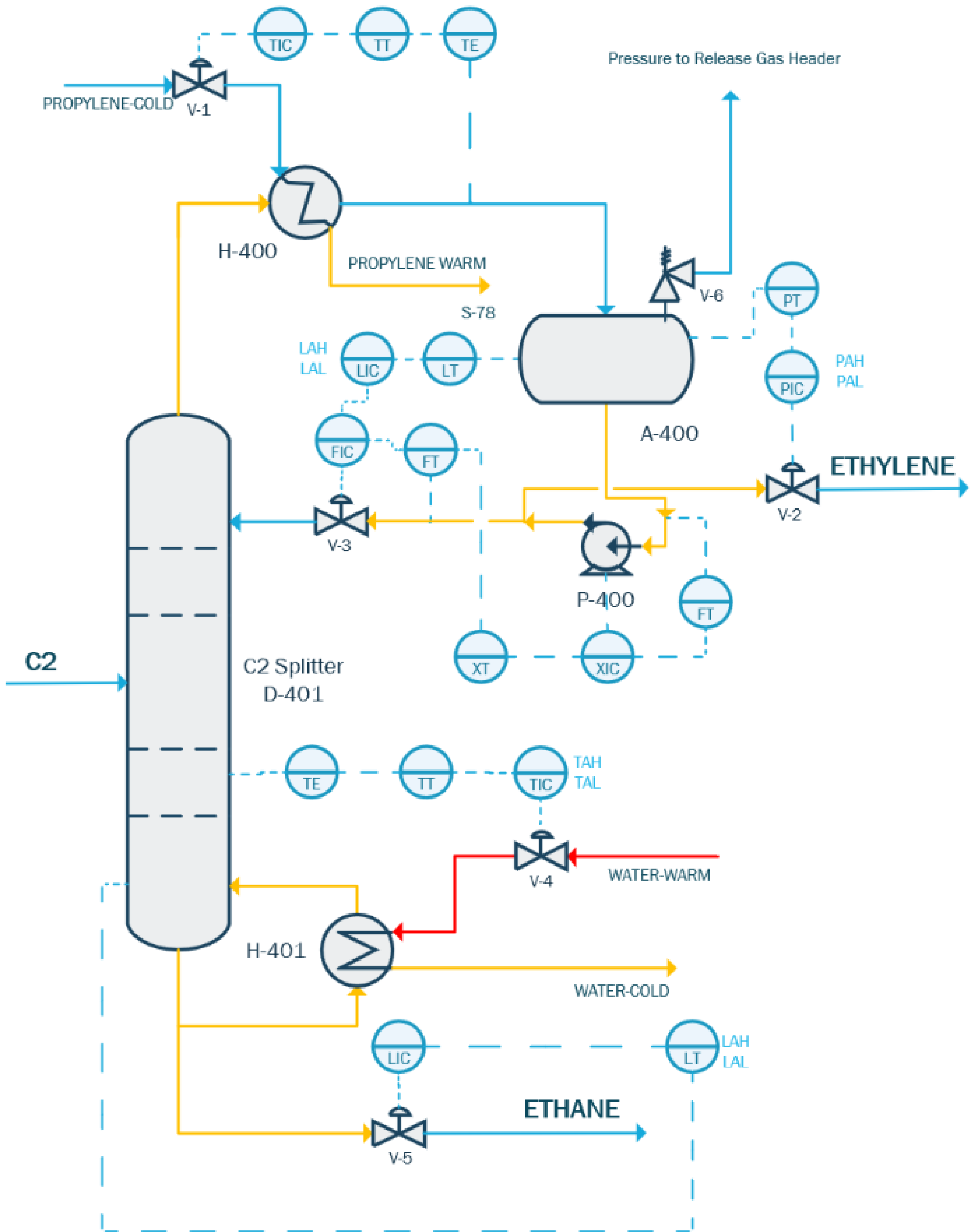

Figure 16: C2 Splitter Overview After HAZOP

When analyzing the ways to optimize the bottom without overwhelming the system with unnecessary controls, it was decided that before HAZOP analysis, the system was efficient. The same control scheme for the bottom will be kept on the C2 splitter but with changes on the top section.

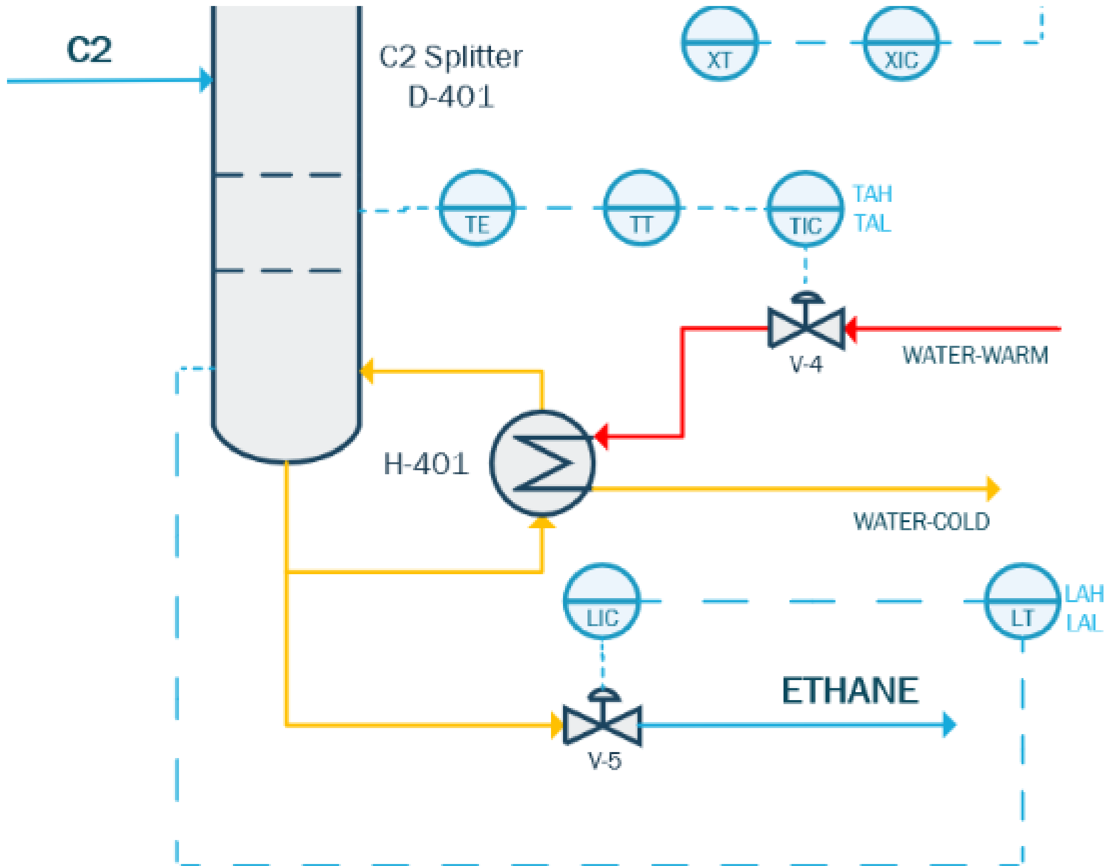

Figure 17: Bottoms After HAZOP

The changes that were introduced at the top section is valve V-6 for pressure control. Instead of being on the column, it is now placed in the accumulator as pressure builds there and it will still control the pressure inside the column D-401 since there is no valve in the previous streams. The spring mechanism will again control the pressure to the same values as that of the column would so only the placement changed. There was also a control system implemented for P-400 in order to prevent reverse flow through the flow transmitters in the inlet and outlet of the pumps. The reason there is a cascade control system for the level in the accumulator and the reboiler ratio is to have control of reverse flow from happening and to ensure the accumulator does not flood. The other control systems in V-1 and V-2 were left the same as it was before.

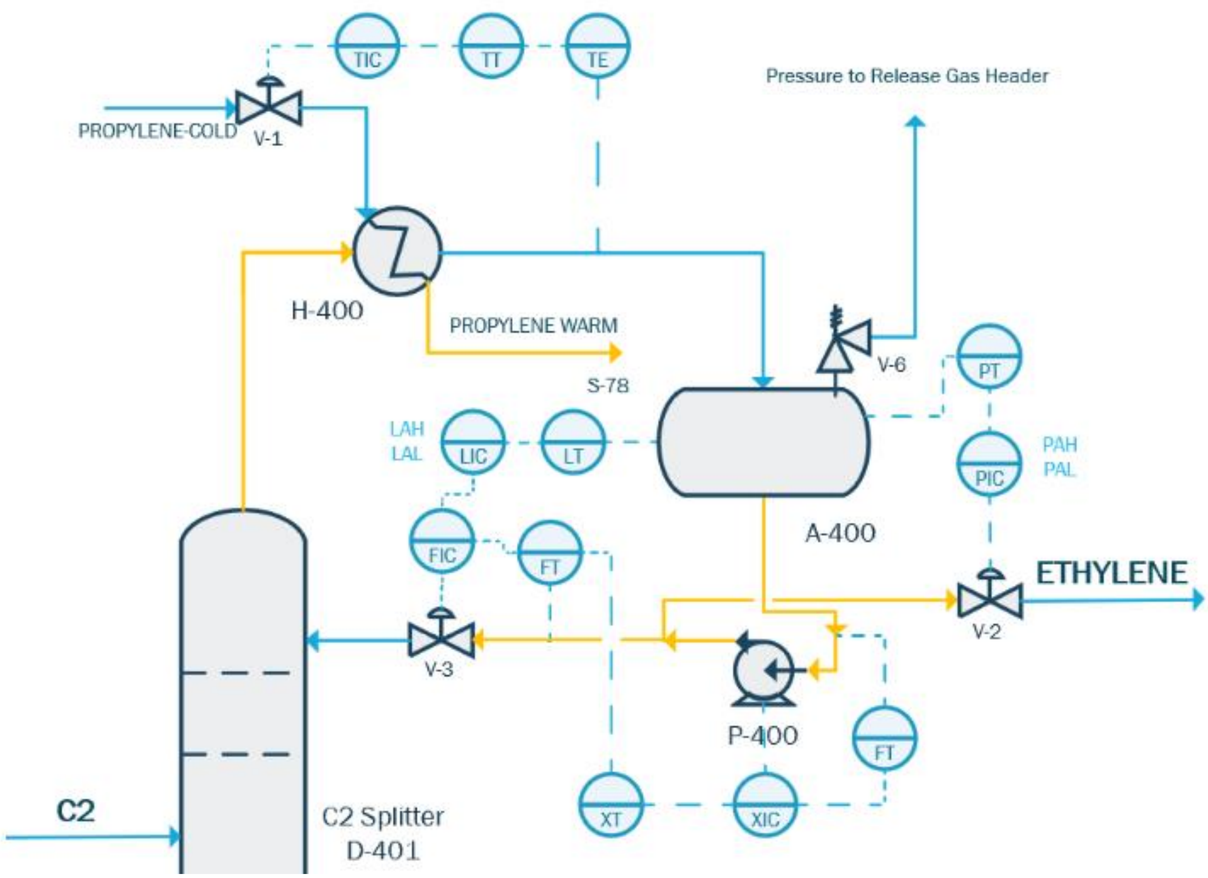

Figure 18: Tops After HAZOP

Safety Analysis

Chemicals

Methanol

Methanol is used in the main reactor as the sole component of the feed stream. Methanol is flammable in both its vapor and liquid form. Methanol is also extremely poisonous and should be handled with care by plant workers. The need for cautious handling is due to the serious health problems that methanol can cause. Methanol is an eye irritant and can cause eye serious damage up to and including blindness.[30] If swallowed, methanol can cause blindness, as well as central nervous system depression.[30] Methanol can cause serious respiratory tract infections if inhaled. Methanol can also cause skin irritation, and if absorbed into the skin can cause other serious effects similar to those of inhalation and ingestion.[30] Due to the serious possible health effects, it's important for all workers to be dressed in PPE. The proper PPE includes chemical splash googles, butyl rubber gloves, and an appropriate respirator.[30] Frequent inspections of all the pipelines and instrumentation where methanol is present is crucial to avoid leaks and accidental inhalation.

The OSHA PEL for methanol is 200 ppm under the time weighted average for an 8-hour shift.[30] The toxic effects of methanol are more serious in humans than animals, and therefore the LD_{LO} orally in humans is 143 mg/kg. The TC_{LO} for methanol in humans is 300 ppm.[30] If methanol is ingested, inhaled, or in contact with the eyes or skin, medical attention is needed. If inhaled, the worker needs to be removed into fresh air immediately.

Environmentally, methanol poses a risk as well. Methanol can be dangerous if it enters the water supply and is very dangerous to aquatic life. The TLM is 96>1000 ppm.[30] However, methanol biodegrades in soil and water rapidly.[30] This process uses methanol as the feed stream, and there is a 100% conversion of methanol to products, therefore disposal considerations are not much of a concern in this case.

The heating of methanol is also quite important due to the autoignition temperature. The autoignition temperature of methanol is 455°C.[30] In this process, the methanol is heated before operating the reactor at 450. The flammability limits of methanol are between 6 and 31 vol percent.[30] Therefore it is crucial to closely monitor the temperature of the methanol stream, as well as the reactor temperature.

Water

Water is one of the products produced from the methanol to olefins reaction. After the reactor, the water is removed. After the water is removed and filtered to take out any remaining olefins, catalyst particles, or other impurities. The impurities are then caked, and the caked material is disposed through an external company. The water is then distilled until it meets the specifications needed to be used as irrigation water. Water is a non-hazardous material, and therefore poses no serious health threats. Water does not normally require any PPE. Frequent inspections of pipelines and instrumentation where water is present are still necessary, in order to protect the equipment and avoid losses.

There is not an OSHA PEL limit listed for water. The LD50 for water is ≥90 mL/kg for a rat.[34]

Environmentally, water is nonhazardous to the environment. However, the water that will be removed from the process needs to be pure enough to be released into the environment. Therefore, the water leaving the plant needs to be incompliance with the US EPA Clean Water Act.

The heating and cooling of water is also not a big concern, as water does not have an autoignition temperature, nor flammability limits.

Ethane

Ethane is a by-product of the methanol to olefins reaction. Ethane is an extremely flammable gas, so caution is required when handling. After the reactor, ethane, a fuel gas, is separated from the light olefin product and either burned at the plant in complex or sent to other refineries. The ethane separated is not released to the environment. Ethane can cause health problems if inhaled or if it comes in contact with the skin or eyes, and in all cases, medical attention should be sought. Liquid ethane can cause frostbite or cold burns.[26] If inhaled, ethane can cause rapid suffocation by displacing oxygen.[26] Workers should wear PPE to prevent any injuries. PPE for ethane handling includes working gloves, safety glasses with side shields, and a respirator when needed.[26] Pipelines and instrumentations should be inspected frequently, in order to prevent any leaks and therefore possible fires or explosions.

There is not a listed OSHA PEL value for ethane. The LC50 for ethane is 658 mg/l in rats.[26] If an individual inhales ethane, the individual needs to be removed from the area, wearing a self-contained breathing apparatus and kept warm.[26] If ethane is in contact with the skin, run the area under warm water. Similarly, if ethane is in contact with the eyes, the eyes need to be flushed with water.[26]

Environmentally, ethane poses no threat. The threat of water and soil pollution is very improbable due to ethane's high volatility. Any waste disposal should follow OSHA regulations.

The heating of ethane should be performed with caution since ethane is highly flammable. The autoignition temperature for ethane is 515°C, which is a temperature that this process will not reach.[26] Although this process will not reach a temperature near that of the autoignition temperature, it is still crucial to closely monitor the temperature of the streams and equipment in which ethane is present.

Propane

Propane is also a by-product of the methanol to olefins reaction. Propane is separated from the final light olefin products and is either used in the plant to generate steam or electricity or is sold. Propane is an extremely flammable gas, and therefore it needs to be handled with caution. Propane can cause health problems if inhaled or if it comes in contact with the skin or eyes, and in all cases, medical attention should be sought. Liquid propane can cause frostbite or cold burns.[34] If inhaled, propane can cause rapid suffocation by displacing oxygen.[34] Workers should wear PPE to prevent any injuries. PPE for propane handling includes safety glasses with side shields, cold insulating gloves, and a respirator when needed.[34] Pipelines and instrumentations should be inspected frequently, in order to prevent any leaks and therefore possible fires or explosions.

The OSHA PEL for propane is 1000 ppm for a time weighted average of an 8 hour shift.[34] The LC50 for propane in a rat is >800000 ppm, with an exposure time of 15 minutes.[34] Therefore, it is crucial to monitor the for propane inhalation in humans. Should an individual inhale propane, they should be removed to fresh air and kept in a comfortable breathing position, and a medical advice should be sought.[34] If propane is in contact with the skin or eyes, flush with water for at least 15 minutes, and seek immediate medical treatment.[34]

Propane poses no environmental hazards, as it is biodegradable, and has a high volatility which prevents water or soil pollution.[34]

The heating of propane should be performed with caution. The autoignition temperature for propane is 450°C, which is a temperature that this process will reach.[34] As some of the equipment and streams in this process reach 450°C, the streams and equipment with propane need to be closely monitored.

Ethylene

Ethylene is the one of the main products in this process. Ethylene is extremely flammable gas, and toxic if inhaled or in contact with skin or eyes. As ethylene is extremely flammable and can cause serious health problems, it requires that it be handled with caution. Possible health problems can occur if inhaled or if ethylene comes in contact with the skin or eyes, and in all cases, medical attention should be sought. Liquid ethylene can cause frostbite or cold burns.[27] If inhaled, ethylene can cause rapid suffocation by displacing oxygen. Workers should wear PPE to prevent any injuries. PPE for ethylene handling includes working gloves, safety glasses with side shields, face shield or vapor proof googles when in contact with the product, and a respirator when needed. [27] Pipelines and instrumentations should be frequently inspected, in order to prevent any leaks and therefore possible fires or explosions.

There is not a listed value for the OSHA PEL for ethylene. There is not a listed LC50 value for ethylene. However, if an individual inhales ethylene, they should be removed to fresh air and medical advice should be sought. [27] If ethylene is in contact with the skin or eyes, flush with water for at least 15 minutes, and seek immediate medical treatment. [27]

Environmentally, ethylene is non-hazardous, as it is biodegradable, so it will not persist. Ethylene has a high volatility which normally prevents water or soil pollution. [27]

The heating of ethylene should be performed with caution. The autoignition temperature for ethylene is 450°C, which is a temperature that this process will reach. [27] As some of the equipment and streams in this process reach 450°C, the streams and equipment with ethylene present need to be closely monitored and controlled properly.

Propylene

Propylene is one of the other main products of this process. Propylene is extremely flammable and can be toxic if inhaled or if it comes in contact with the skin or eyes. Due to this, propylene needs to be handled with care, to avoid any unnecessary health problems. The potential health hazards can occur if inhaled or if propylene comes in contact with the skin or eyes, and in all cases, medical attention should be sought. Liquid propylene can cause frostbite or cold burns. [33] If inhaled, propylene can cause rapid suffocation by displacing oxygen.[33] Workers should wear PPE to prevent any injuries. PPE for propylene handling includes working gloves or welding gloves if welding, safety glasses with side shields, and a respirator when needed. [33] Pipelines and instrumentations should be frequently inspected, in order to prevent any leaks and therefore possible fires or explosions.

There is not a listed value for the OSHA PEL for propylene. The LC50 for propylene in a rat is >65000 ppm.[33] If an individual inhales propylene, they should be removed to fresh air and medical advice should be sought. [33] If propylene is in contact with the skin or eyes, flush with water for at least 15 minutes, and seek immediate medical treatment. [33]

Environmentally, propylene is non-hazardous, as it is biodegradable, so it will not persist. Propylene has a high volatility which normally prevents water or soil pollution. [33]

The heating of propylene should be performed with caution. The autoignition temperature for propylene is 455℃, which is a temperature that this process will come close to. [33] As some of the equipment and streams in this process near 455℃, the streams and equipment with propylene present need to be closely monitored and controlled properly.

Butene

Butene is one of the by-products produced in this process. Butene is an extremely flammable gas and can cause serious health complications. The health hazards can occur if inhaled or if butene comes in contact with the skin or eyes, and in all cases, medical attention should be sought. [23] If inhaled, butene can cause rapid suffocation by displacing oxygen.[23] Workers should wear PPE to prevent any injuries. PPE for butene handling includes working gloves, safety glasses with side shields, metatarsal shoes, and a respirator when needed. [23]

Pipelines and instrumentations should be frequently inspected, in order to prevent any leaks and therefore possible fires or explosions.

There is not a listed value for the OSHA PEL for butene. There is not a listed LC50 value for butene. However, if an individual inhales butene, they should be removed to fresh air and medical advice should be sought.[23] If butene is in contact with the skin or eyes, flush with water for at least 15 minutes, and seek immediate medical treatment. [23]

Butene is not biodegradable, so it will persist in the environment, but it is not hazardous to the environment. butene has a high volatility, which prevents water or soil pollution. [23]

The heating of butene should be performed with caution. The autoignition temperature for butene is 384℃, which is a temperature that this process will reach in certain areas. [23] As some of the equipment and streams in this process reach 384℃, the streams and equipment with butene present need to be closely monitored and controlled properly.

Dimethyl Ether

Dimethyl Ether is an intermediate product that is present in this process. Any Dimethyl ether that is produced in the methanol to olefins reaction is separated and completely recycled back to the reactor. Dimethyl ether is an extremely flammable gas and is toxic. The health hazards can occur if dimethyl ether is inhaled or if dimethyl ether comes in contact with the skin or eyes, and in all cases, medical attention should be sought.[25] If inhaled, dimethyl ether can cause rapid suffocation by displacing oxygen, as well as drowsiness or dizziness. [25] Liquid dimethyl ether can cause frostbite or cold burns. Workers should wear PPE to prevent any injuries. PPE for dimethyl ether handling includes working gloves, safety glasses with side shields, and a respirator when needed.[25] Pipelines and instrumentations should be frequently inspected, in order to prevent any leaks and therefore possible fires or explosions.

There is not a listed value for the OSHA PEL for dimethyl ether. The LC50 for dimethyl ether in a rat is 163,754 ppm. [25] If an individual inhales dimethyl ether, they should be removed to fresh air and medical advice should be sought. [25] If dimethyl ether is in contact with the skin or eyes, flush with water for at least 15 minutes, and seek immediate medical treatment. [25]

Dimethyl ether is not biodegradable, so it will persist in the environment, but it is not hazardous to the environment. Dimethyl ether has a high volatility, which prevents water or soil pollution. [25] Dimethyl ether can cause slight pH changes in aquatic ecological systems and has potential to effect on global warming. [25] In this process, dimethyl ether is used as an intermediate product, and the dimethyl ether that is separated out is recycled back into the reactor.

The heating of dimethyl ether should be performed with extreme caution. The autoignition temperature for dimethyl ether is 350℃, which is a temperature that this process will certainly reach. [25] Therefore, the streams and equipment with dimethyl ether present need to be closely monitored and controlled properly.

Methane

Methane is one of the by-products produced in this process. Methane is an extremely flammable gas, so caution is required when handling. Methane can cause health problems if inhaled and medical attention should be sought. If inhaled, methane can cause rapid suffocation by displacing oxygen. [29] Workers should wear PPE to prevent any injuries. PPE for methane handling includes working gloves, safety glasses with side shields, a face shield or vapor proof googles when in contact with the product, metatarsal shoes, and a respirator when needed. [29] Pipelines and instrumentations should be inspected frequently, in order to prevent any leaks and therefore possible fires or explosions.

There is not a listed OSHA PEL value for methane. There is not a listed LC50 value for methane. However, if an individual inhales methane, they should be removed to fresh air and medical advice should be sought. [29] If methane is in contact with the eyes, flush with water for at least 15 minutes, and seek immediate medical treatment. [29]

Environmentally, methane is slightly hazardous. The threat of water and soil pollution is very improbable due to methane's high volatility. Methane is also biodegradable. However, methane has a high global warming potential, and can contribute to global warming when released in large quantities. [29]

The heating of methane should be performed with caution, since methane is highly flammable. The autoignition temperature for methane is 600℃, which is a temperature that this process will not reach. [29] Although this process will not reach a temperature near that of the autoignition temperature, it is still crucial to closely monitor the temperature of the streams and equipment in which methane is present.

Butane

Butane is one of the by-products produced in this process. Butane is an extremely flammable gas, so caution is required when handling. Butane can cause health problems if inhaled or if butane comes in contact with the skin or eyes, and in all cases medical attention should be sought. If inhaled, butane can cause rapid suffocation by displacing oxygen. [22] If in contact with the skin, frostbite or cold burns can occur. Workers should wear PPE to prevent any injuries. PPE for butane handling includes cold insulating gloves, safety glasses with side shields, and safety shoes. [22] Pipelines and instrumentations should be inspected frequently, in order to prevent any leaks and therefore possible fires or explosions.

There is not a listed OSHA PEL value for butane. The LC50 for butane in a rat is 658 mg/L with an exposure time of 4 hours. [22] However, if an individual inhales butane, they should be removed to fresh air, kept warm and medical advice should be sought. [22] If butane is in contact with the eyes, flush with water for at least 15 minutes, and seek immediate medical treatment. [22] If butane is in contact with the skin, run under warm water for at least 15 minutes and seek medical attention.

Environmentally, butane is not hazardous. The threat of water and soil pollution is very improbable due to butane's high volatility. Butane is also biodegradable. [22]

The heating of butane should be performed with caution, since butane is highly flammable. The autoignition temperature for butane is 400℃, which is a temperature that this process will reach. [22] As certain equipment and streams will operate at or above 400℃ in this process, it is crucial to closely monitor the temperature of the streams and equipment in which butane is present and control those streams properly.

Pentane

Pentane is one of the byproducts produced in this process. Pentane is an extremely flammable gas, so caution is required when handling. Pentane can cause health problems if inhaled or if pentane comes in contact with the skin or eyes, and in all cases medical attention should be sought. If inhaled, pentane can drowsiness and dizziness, as well as damage to the central nervous system and respiratory system. [31] If pentane comes in contact with the skin or eyes, irritation can occur. Workers should wear PPE to prevent any injuries. PPE for pentane handling includes cold insulating gloves, chemical splash goggles, and a respirator when necessary. [31] Pipelines and instrumentations should be inspected frequently, in order to prevent any leaks and therefore possible fires or explosions.

The OSHA PEL for pentane is 1000 ppm for a time weighted average of an 8-hour shift.[31] The LC50 for pentane in a rat is 280000 mg/m^3, with an exposure time of 4 hours. [31] If an individual inhales pentane, they should be removed to fresh air and medical advice should be sought. [31] If pentane is in contact with the skin or eyes, flush with water for at least 15 minutes, and seek immediate medical treatment. [31]

Environmentally, pentane is not hazardous. The threat of water and soil pollution is very improbable due to pentane's high volatility. Pentane is also biodegradable. [31]

The heating of pentane should be performed with caution since pentane is highly flammable. The autoignition temperature for pentane is 260°C, which is a temperature that this process will reach. [31] As certain equipment and streams will operate at or above 260°C in this process, it is crucial to closely monitor the temperature of the streams and equipment in which pentane is present and control those streams properly.

Hydrogen

Hydrogen is one of the by-products produced in this process. Hydrogen is an extremely flammable gas, so caution is required when handling. Hydrogen can cause health problems if inhaled or if hydrogen comes in contact with the eyes, and in all cases medical attention should be sought. If inhaled, can cause rapid suffocation due to oxygen displacement. [28] If in contact with the eyes, eye damage can occur. Workers should wear PPE to prevent any injuries. PPE for hydrogen handling includes safety glasses with side shields and a respirator when necessary. [28] Pipelines and instrumentations should be inspected frequently, in order to prevent any leaks and therefore possible fires or explosions.

The OSHA PEL for hydrogen is not listed. [28] The LC50 for hydrogen in a rat is 15000 ppm, with an exposure time of 1 hour. [28] If an individual inhales hydrogen, they should be removed to fresh air and medical advice should be sought. [28] If hydrogen is in contact with the eyes, flush with water for at least 15 minutes, and seek immediate medical treatment. [28]

Environmentally, hydrogen is not hazardous and does not pose a threat.

The heating of Hydrogen should be performed with caution since hydrogen is highly flammable. The autoignition temperature for hydrogen is 566°C, which is a temperature that this process will not reach. [28] Although this process will not reach 566°C, it is still crucial to closely monitor the temperature of the streams and equipment in which hydrogen is present and control those streams properly.

Carbon Dioxide

Carbon dioxide is present as it is one of the carbon oxides released from the catalyst regenerator. Carbon dioxide can act as an asphyxiant, so caution is required when handling. Carbon dioxide can cause health problems if inhaled or if it comes in contact with the skin or eyes, and in all cases medical attention should be sought. If inhaled, can cause rapid suffocation due to oxygen displacement, as well as frostbite. [24] If in contact with the skin, frostbite or cold burns can occur. If in contact with the eyes, eye damage can occur. [24] Workers should wear PPE to prevent any injuries. PPE for Carbon dioxide handling includes work gloves, heavy rubber gloves where direct contact may occur, safety glasses with side shields, a face shield when in contact with the product and a respirator when necessary. [24] Pipelines and instrumentations should be inspected frequently, in order to prevent any leaks.

The OSHA PEL for carbon dioxide is 5000 ppm for the time weighted average of an 8-hour shift. [24] The LC50 for carbon dioxide is not listed. If an individual inhales carbon dioxide, they should be removed to fresh air, kept warm, and medical advice should be sought. [24] If carbon dioxide is in contact with the eyes, flush with water for at least 15 minutes, and seek immediate medical treatment. [24] If carbon dioxide is in contact with the skin, the area should be run under warm water for at least 15 minutes, and medical advice is needed.

Environmentally, carbon dioxide has a role in global warming. Carbon Dioxide has a small global warming potential and can contribute a lot of the greenhouse effect when released in large quantities. Carbon dioxide can also cause frost damage to crops. [24]

Carbon dioxide does not have an autoignition temperature, so the heating of carbon dioxide is not a direct cause for concern. [24]

Carbon Monoxide

Carbon monoxide is present in the methanol to olefins process. Carbon monoxide is present in the cold box, with the DME, in the fuel gas in extremely small quantities, and as one of the carbon oxides released from the catalyst regenerator. After the carbon monoxide is burned, it is released in the air as ppm of CO_2. Carbon monoxide an extremely flammable gas and is toxic if inhaled. Carbon monoxide can cause damage to the central nervous system through repeated or prolonged exposure, as well as fertility issues, and damage to unborn children.[38] Carbon monoxide acts as an asphyxiant, even with enough oxygen. Therefore, carbon monoxide needs to be handled with caution and PPE is needed. The PPE for carbon monoxide includes working gloves, safety glasses, a face shield when needed, and a respirator when working in a small area or an area with inadequate ventilation.[38]

The US OSHA PEL limit for carbon monoxide is 50 ppm for the time weighted average of an 8 hour shift.[38] The LC50 in a rat is 1880 ppm with an exposure time of 4 hours.[38} If an individual inhales carbon monoxide, they should be removed to fresh air, and should seek medical attention. If the individual is not breathing, the airways should be cleared, and artificial respiration should be given. If carbon monoxide comes in contact with the skin or water, they should be flushed with water for at least 15 minutes, and medical advice is needed.

Environmentally, carbon monoxide is not hazardous. Carbon monoxide does not cause ecological damage, nor will it cause water or ground pollution due to its high volatility. [3??} However, carbon monoxide does have a potential to effect global warming. Carbon monoxide is not readily biodegradable.

Although carbon monoxide does not have a large presence in this process, it is extremely flammable. Carbon monoxide has an auto ignition temperature of 605°C, which is a temperature that this process will not reach. [38] Although this process will not reach 605°C, it is important to monitor the temperature in areas where carbon monoxide is present.

Table 8 - Hazardous Chemical Properties [22-34,38]

Chemical	Boiling point	Flash point	Auto-ignition	Flammability
Methanol	64.7°C	12°C	455°C	Upper: 31 vol% Lower: 6 vol %
Water	100°C	N/a	N/a	N/a
Ethane	-88°C	-135°C	515°C	Upper: 12.5 vol% Lower: 3 vol%
Propane	-42.1°C	-104.4°C	450°C	Upper 9.5 vol% lower 2.1 vol%
Ethylene	-102.4°C	-136.1°C	450°C	Upper 36 vol% Lower 2.7 vol%
Propylene	-47.7°C	-107.8°C	455°C	Upper 11.1 vol% Lower 2 vol%
Butene	-6.47°C	-110 F	384°C	Upper 9.3 vol% Lower 1.6 vol%
Dimethyl ether	-24.8°C	N/a	350°C	Upper 18 vol% Lower 3.4 vol %
Methane	-161.5°C	N/a	600°C	Upper 15 vol% Lower 5 vol%
Butane	-0.5°C	-60°C	400°C	N/a
Pentane	36°C	-49°C	260°C	Upper 7.8 vol% Lower 1.5 vol%
Hydrogen	-252.9°C	N/a	566°C	N/a
Carbon Dioxide	-78.4°C	N/a	N/a	N/a
Carbon Monoxide	-191.5°C	N/a	605°C	Upper: 74 vol% Lower: 12.5 vol%

Operational Safety:

Table 9: Operational Safety [36]

Process	Main Safety Concern	Solution
Preheating Methanol	1. There is a risk of an explosion if there is a leak in the pipeline or instrumentation due to the pressure and temperature of the methanol stream. 2. The fluctuation of the methanol flow into the reactor can cause the reaction temperature to be greater than what it needs to be.	1. Frequently inspect all pipelines, pipeline connectors, and instrumentation for possible leaks. 2. Control the flow of the methanol stream from the preheating unit to the reactor unit
Reactor	1. The MTO reaction is extremely exothermic, therefore there are risks involving the temperature of the reactor and the overheating of the reactor. 2. There is a risk of an explosion if there is a leak in the fluidized be reactor	1. Control the temperature of the methanol stream from the preheating unit to the reactor. 2. Frequently inspect all pipelines surrounding the reactor and the reactor itself and perform routine maintenance.
Catalyst Regenerator	1. The is a risk of an explosion if the air from the catalyst regeneration makes its way into the reaction and comes in contact with the gas phase.	1. Ensure that the pressure balance between the reactor and the regenerator and the regenerator. Also perform routine maintenance.
Quenching Unit	1. Variations in the liquid level could cause the feed pipeline to become plugged, the tower to flood, and water to back-flow into the system.	1. Control the level of the liquids in the towers in the quench hierarchy.
Separation units	1. There is a risk of the temperature rising above the needed temperature.	1. Control the temperatures of the inlet streams to each separation unit.

Environmental Safety:

Water discharge and air emissions are crucial factors when considering the impact that a process has on the environment. Although this plant is located on the west coast of India, the environmental standards will follow the US EPA guidelines.

The advanced MTO process produces water as a product, and that water is later discharged. The water streams that are discharged from this process only consist of water, due to the high product specification requirement. Therefore, there are no added chemicals in the discharge streams of this process. After the water in this process is removed, it is filtered, caked, and distilled until it meets the specification standards to be used as irrigation in areas neighboring the plant.

The catalyst regenerator is the source of air emissions. As the regenerator is removing the coke build-up from the catalysts, the flue gas leaving the regenerator consists of carbon oxides. These carbon oxides can be harmful to the environment, as they contribute to the greenhouse effect. Therefore, the flue gas cannot simply be released into the atmosphere, and need to be treated in accordance with the US EPA's Clean Air Act.[37] The flue gas leaving the catalyst regeneration needs to be treated using a post combustion CO_2 capture.[35] The capture of CO_2 can be achieved by using amine absorption, as it is quite successful in removing the CO_2.[35] Once the flue gas has been treated, the air emissions will be in compliance with the US EPA regulations.

HAZOP Analysis:

Table 10: HAZOP Analysis of Distillation Column (C2 Splitter)

Vessel	C2 Splitter: S-401				
Intention	Separates Ethylene and Ethane				
Guide word	Deviation	Cause	Consequences	Safeguards	Action
Stream	C2				
Intention	Provides feed stream to the C2 Splitter				
No/Less	Flow	Failure in S-400 reboiler, Blocked pipes	Low level in S-401	Add low level alarm on S-401	Perform maintenance on S-400 reboiler, remove blockage
More	Flow	Failure in S-400 temperature control	High level in S-401	Add high level alarm on S-401	Perform maintenance on control instrumentation
As Well As	Composition	Failure in in S-400 separation	Product streams do not meet specifications	Add concentration control loop	Perform maintenance on S-400
Stream	Ethylene				
Intention	Product stream leaving distillation column				

No/Less	Flow	Failure in S-401 reboiler, blocked pipes	Loss of product stream, ethane stream does not meet specifications.	Add temperature alarm on S-401	Remove pipe blockage. Perform maintenance on S-401 reboiler.
More	Flow	Failure in temperature control	Loss of product stream, ethylene stream does not meet specifications.	Add temperature alarm on S-401	Perform maintenance on control instrumentation
As well As	Composition	Failure in S-401 Separation	Product stream does not meet specifications	Add concentration control loop	Perform maintenance on S-401
Stream	Ethane				
Intention	Product stream leaving distillation column				
No/Less	Flow	Failure in temperature control	Loss of product stream, ethylene stream does not meet specifications.	Add temperature alarm on S-401	Perform maintenance on control instrumentation
More	Flow	Failure in reboiler	Loss of product stream, ethane stream does not meet specifications	Add temperature alarm on S-401	Remove pipe blockage. Perform maintenance on S-401 reboiler.
As Well As	Composition	Failure in S-401 separation	Product stream does not meet specifications	Add concentration control loop	Perform maintenance on S-401
Column	S-401				
Intention	Separates Ethylene and Ethane				
Low	Level	Failure in level control	Damage to equipment	Add low level alarm on	Perform maintenance on

				S-401	control instrumentation and S-400 reboiler
More	Level	Failure in level control	Equipment Flooding	Add high level alarm on S-401	Perform maintenance on control instrumentation

Economic Analysis:

Equipment Cost:

Using table 7.2, in the Gavin Towler senior design book the equipment costs were obtained. From this table, four different values were obtained: a, b, n, and size values. These obtained values were inputted into the following equation to get the price of each piece of equipment [39]. The Lang factor was a factor the was multiplied which depended on the type of equipment you had when doing this calculation. Lastly the total equipment cost was multiplied by 5 to account for the installation cost.

$$Cost\ of\ equipment = (\mathrm{a} + \mathrm{b}\mathrm{S}^{n}) * Lang\ Factor * Installation\ Cost \qquad (1)$$

Lastly, after obtaining the total major equipment cost, this number had to be converted to the 2019 index because the table used only provides the numbers in the 2010 index. Using the flowing equation, we obtained our total major equipment cost in the 2019 cost index.

$$Cost\ Index_{2019} = Cost\ Index_{2010} * \left(\frac{Location\ Index\ 2019}{Location\ Index\ 2010}\right) * \left(\frac{Cost\ Index\ 2019}{Cost\ Index\ 2010}\right) \qquad (2)$$

Table 11: Total Major Equipment Costs

Equipment	Amount	Material	Type of equipment	a	b	n	Size	Units	Cost
Reactor (R-100)	1	Carbon Steel	R-Yield	61500	32500	0.8	190.6	m^2	$ 5,572,781.64
Reactor (R-101	1	Carbon Steel	R-Yield	61500	32500	0.8	21.45	m^2	$ 1,097,735.25
Reactor (R-400)	1	Carbon Steel	R-Yield	61500	32500	0.8	19.06	m^2	$ 1,012,608.64
Heat Exchanger (H-100)	1	Carbon Steel	Shell and Tube Heat Exchanger	28000	54	1.2	265.501	m^2	$ 251,229.08
Heat Exchanger (H-200)	1	Carbon Steel	Shell and Tube Heat Exchanger	28000	54	1.2	5.37858	m^2	$ 99,423.19
Heat Exchanger (H-300)	1	Carbon Steel	Shell and Tube Heat Exchanger	28000	54	1.2	10.9417	m^2	$ 101,337.06
Heat Exchanger (H-400)	1	Carbon Steel	Shell and Tube Heat Exchanger	28000	54	1.2	0.63536	m^2	$ 98,109.67
Heat Exchanger (H-401)	1	Carbon Steel	Shell and Tube Heat Exchanger	28000	54	1.2	50.47	m^2	$ 118,897.86
Compressor (C-300)	1	Carbon Steel	Centrifugal compressor	580000	20000	0.6	7394.54	kW	$ 11,928,924.27
DeEthanizer Column	1	Carbon Steel	Vertical Pressure Vessel	11600	34	0.85	123467.8	kg	$ 1,837,171.87
DeEthanizer Reboiler	1	Carbon Steel	Thermosiphen Reboiler	30400	122	1.1	9.63957	m^2	$ 79,687.77
DeEthanizer Condenser	1	Carbon Steel	U-Tube Shell and Tube	28000	54	1.2	3350.61	m^2	$ 2,363,413.82
DePropanizer Column	1	Carbon Steel	Vertical Pressure Vessel	11600	34	0.85	173181.6	kg	$ 2,439,712.06
DePropanizer Reboiler	1	Carbon Steel	Thermosiphen Reboiler	30400	122	1.1	52.057	m^2	$ 99,573.65
DePropanizer Condenser	1	Carbon Steel	U-Tube Shell and Tube	28000	54	1.2	39.23818	m^2	$ 81,035.31
C2-C3 Column	1	Carbon Steel	Vertical Pressure Vessel	11600	34	0.85	146655.5	kg	$ 2,122,015.72
C2-C3 Reboiler	1	Carbon Steel	Thermosiphen Reboiler	30400	122	1.1	27.80456	m^2	$ 87,825.72
C2-C3 Condenser	1	Carbon Steel	U-Tube Shell and Tube	28000	54	1.2	12310.91	m^2	$ 11,001,557.24
DeMethanizer Column	1	Carbon Steel	Vertical Pressure Vessel	11600	34	0.85	121677.8	kg	$ 1,814,865.04
DeMethanizer Reboiler	1	Carbon Steel	Thermosiphen Reboiler	30400	122	1.1	8.3649	m^2	$ 79,155.05
DeMethanizer Condenser	1	Carbon Steel	U-Tube Shell and Tube	28000	54	1.2	55.876	m^2	$ 86,865.69
Pump	1	Carbon Steel	Single Stage centrifugal	8000	240	0.9	71.46	l/s	$ 76,763.90
Pump	1	Carbon Steel	Single Stage centrifugal	8000	240	0.9	28.18	l/s	$ 51,373.95
Pump	1	Carbon Steel	Single Stage centrifugal	8000	240	0.9	32.3	l/s	$ 53,905.52
Pump	1	Carbon Steel	Single Stage centrifugal	8000	240	0.9	32.09	l/s	$ 53,777.30
Pump	1	Carbon Steel	Single Stage centrifugal	8000	240	0.9	30.87	l/s	$ 53,030.72
Pump	1	Carbon Steel	Single Stage centrifugal	8000	240	0.9	0.87	l/s	$ 32,846.91
Pump	1	Carbon Steel	Single Stage centrifugal	8000	240	0.9	1.12	l/s	$ 33,063.08
Pump	1	Carbon Steel	Single Stage centrifugal	8000	240	0.9	29.74	l/s	$ 52,336.59
Pump	1	Carbon Steel	Single Stage centrifugal	8000	240	0.9	22.4	l/s	$ 47,757.76
Pump	1	Carbon Steel	Single Stage centrifugal	8000	240	0.9	21.54	l/s	$ 47,212.21
Pump	1	Carbon Steel	Single Stage centrifugal	8000	240	0.9	16.8	l/s	$ 44,163.25
Pump	1	Carbon Steel	Single Stage centrifugal	8000	240	0.9	5.6	l/s	$ 36,525.23
Pump	1	Carbon Steel	Single Stage centrifugal	8000	240	0.9	7.2	l/s	$ 37,673.75
Pump	1	Carbon Steel	Single Stage centrifugal	8000	240	0.9	14.34	l/s	$ 42,547.89
Pump	1	Carbon Steel	Single Stage centrifugal	8000	240	0.9	14.19	l/s	$ 42,448.53
Pump	1	Carbon Steel	Single Stage centrifugal	8000	240	0.9	0.14	l/s	$ 32,163.60

Total Major Equipment Cost (2019) = $242.33 MM

Comparison of Milos Calculator to Aspen Cost:

Our total equipment cost using the Milos Calculator which is based off the Gavin Towler book was about $242.33 million. Running our Aspen Simulation gave a total of equipment plus installed cost of 19 million. The Aspen Simulation was much lower than using the table in the senior design book for the equipment cost. However, the rest of the calculations were made using the Gavin Towler book estimation and the Milos Calculator.

ISBL:

This cost was assumed to be equal to the major equipment cost obtained above. This is because it included the cost of each piece of equipment in the process and the cost to install it, which is covered in the above example.

OSBL:

The offsite investment cost which is the OSBL cost includes the cost of any additions to the original site infrastructure typically to increase the capacity of the existing plant [39]. This includes anything from boilers, cooling towers, water pipes to extra offices, laboratories, and emergency services [39]. In this analysis the assumption was made that this cost is 40% of the ISBL cost.

$$OSBL\ Cost = 40\% \times ISBL\ Cost \tag{3}$$

Fixed Capital Cost:

The fixed capital cost is just the additions of the ISBL and OSBL costs. These costs were stated and explained above.

$$Fixed\ Capital\ Cost = ISBL + OSBL \tag{4}$$

Working Capital:

The working capital is simply the capital which has to be invested into the plant in order to make sure the operations of the plant are maintained. This capital is what reassures that the inventory of your feed and products is maintained [39]. However, is the plant shuts down, this capital is returned to the investor since it's not needed anymore to maintain operations [39]. Typically, the working capital is assumed to be 10% of the ISBL cost.

$$Working\ Capital = 10\% \times ISBL\ Cost \tag{5}$$

Contingency:

This cost simply is any extra cost that should be added to the budget of designing the plant to cover anything from minor changes to labor disputes that are unforeseen [39]. This is typically is assumed to be 10% of the ISBL cost.

$$Contingency = 10\% \times ISBL\ Cost \tag{6}$$

Engineering Cost:

Any engineering services that are needed to design the plant from design engineering of process equipment to the profit of the contractor onsite is considered the engineering cost [39]. This is typically is assumed to be 10-30% of the ISBL cost.

$$Engineering\ Cost = \frac{10\% \ * \ \text{ISBL Cost} + \ 30\% \ * \ \text{ISBL Cost}}{2} \tag{7}$$

Total Capital Cost:

Lastly the total capital cost is the addition of all the previous costs stated above. The following equation is:

Total Capital Cost = ISBL+ OSBL+ Working Capital+ Contingency+ Engineering cost

Table 12: Total Capital Cost

Total Capital Cost($MM/year)		
ISBL/Major Equipment Cost	$	242.33
OSBL	$	96.93
Fixed Capital Cost	$	339.36
Working Capital	$	24.23
Contingency	$	24.23
Engineering Cost	$	48.47
Initial Cost of SAPO-34 Catalyst	$	15.00
Waste Disposal Annaully	$	10.00
Total Capital Cost	$	461.19

Variable Cost of Production:

This cost includes the addition of the total cost of raw materials and the total cost of utilities. In our process we have one raw material which is methanol. We will be using 1,623,000 MT/year of methanol. Due to the investment in a methanol plant in Angola and the contract between us and this country, we will be able to obtain methanol at cost of 215 dollars/ton, which would be less than all our competitors [40]. This is because the methanol plant in Angola will only ship the methanol to us and there will be no other customers. Knowing how much will need and the cost of raw material, gives us a total cost of raw materials at $401.38 MM/ year. Next, looking at the total cost of utilities, we have electricity, cooling water, steam, and our catalyst. In table 14, we see how much is needed per year in our process and at what price it will cost us. Adding up all these costs give a total cost of utilities at $20.5 MM/year. The following equation gives us our variable cost of production.

$$VCOP = Total\ Cost\ of\ Raw\ Materials + Total\ Cost\ of\ Utilities \tag{8}$$

Table 13: Raw Material Cost

Total Cost of Raw Materials				
Raw Materials	Quantity	Total	Cost	Total Cost/year ($MM/year)
Methanol	1623000 MT/year	1472360.8 ton/year	$215/ton	$ 347.50
SAPO-34 Catalyst	39.6 ton/year	22679 kg/year	$150/kg	$ 5.38
Methanol Transportaion	1466267.69 MT/year	1616283.46 ton/year	$30/ton	$ 48.49

Table 14: Utility Cost and Total Cost

Utilities	Quantity	Cost	Total Cost/year $(MM/year)
Electricity	2488000 kW/year	5.996/kW	$ 14.90
Cooling Water	1400 Mt/hr	0.5/Mt	$ 5.60
Steam	1524.098 Mt/hr	0.004/Pound	$ 0.83
Propylene	5.38 Mt/hr	Used From Product	
Ethylene	30.93 Mt/hr	Used From Product	

Total Cost of Raw Materials	**$ 401.38**
Total Cost of Utilities	**$ 20.56**

Fixed cost of Production:

Our plant will operate 24 hours/day for 330 days which is 7920 hours/year. This plant will run on a 4 shifts a day schedule, with 5 operators per shift position. We assume each operator will be getting $60,000/year [43]. Using the excel template from the Gavin Towler book and inputting the above information, we calculated based off given percentages the basic salary overhead, maintenance, general plant overhead, tax and insurance. Knowing all these variables we obtained the fixed cost of production to be $33.76 MM/year.

Operating Variables				Description			$M/yr $M
Labor							
Operators per Shift Position	4.8						
Number of shift positions	4		$60,000.00	$/yr each			$1,152.00
Supervision			25%	of Operating Labor			$288.00
Direct Ovhd.			45%	of Labor & Superv.	$2,088.00		$648.00
Maintenance			3%	of ISBL Investment			$13,085.61
Overhead Expense							
Plant Overhead			65%	of Labor & Maint.	$18,586.59		$9,862.85
Tax & Insurance			2%	of Fixed Investment			$8,723.74
Interest on Debt Financing			0%	of Fixed Capital			$0.00
Type1			0%	of Working Capital			$0.00
Type2							
							$33,760.20

Figure 19: Fixed Cost of Production

Revenues and Gross Profit:

Our revenue will be generated from our two main products of ethylene and propylene. In table 15, we see how much of our products we are making per year. We can also see that we will be selling ethylene at $935/MT and propylene at $825/MT [42]. Knowing this, our total main products revenue will $609.7MM/year. Lastly our Gross Profit will be our total main product revenue minus our variable and fixed cost of production. This is shown in the following equation:

$$Gross\ Profit = Main\ Product\ Revenue - (VCOP - FCOP) \tag{9}$$

Table 15: Revenues and Gross Profit

Revenues	How much are we making	Unit	Cost($)	Unit	Revenue ($MM/year)
Ethylene	597,447	MT/year	935	MT	$ 558.61
Propylene	61895	MT/year	825	MT	$ 51.06

Gross Profit ($MM/year)
$ 153.98

Pay Back Time:

Our payback time is the time it will take to get back our original investments into our plant. This will be determined by the average annual cash flow (Gross Profit) divided by our total capital cost investment as shown in the following equation:

$$Pay\ Back\ Time = \frac{Gross\ Profit}{Total\ Capital\ Cost} \tag{10}$$

Table 16 – Pay Back Time

Pay Back Time	
Average Annual Cash Flow	$ 153.98
Total Investment	$ 461.18
Time in years	2.995064294

Return of investment:

Similar to the payback time above, this will be determined by our total capital cost investment divided by the average annual cash flow (Gross Profit) as shown in the following equation.

$$Return\ of\ Investment = \frac{Total\ Capital\ Cost}{Gross\ Profit} \tag{11}$$

Table 17: ROI

ROI	
Net Annual Profit	$ 153.98
Total investment	$ 461.18
Pre Tax ROI	33.38826489

NPV Analysis:

Inputting all our above values into Milos's calculator the NPV table was obtained. We assumed it will take 2 years of construction period and a 15.65% interest rate and did an NPV table for a 5- and 7-year depreciation period with the MACRS depreciation method. The tables are shown below with the last number being our NPV value at each depreciation period.

Table 18: 5 Year NPV

Project year	Cap Ex	Revenue	CCOP	Gr. Profit	Deprcn	Taxbl Inc	Tax Paid	Cash Flow	PV of CF	NPV
1	138.4	0.0	0.0	0.00	0.00	0.0	0.0	-138.4	-119.6	-119.6
2	322.8	0.0	0.0	0.00	0.00	0.0	0.0	-322.8	-241.4	-361.0
3	0.0	0.0	0.0	0.00	92.24	-92.2	0.0	0.0	0.0	-361.0
4	0.0	0.0	0.0	0.00	147.58	-147.6	0.0	0.0	0.0	-361.0
5	24.2	304.8	244.7	60.11	88.55	-28.4	0.0	35.9	17.3	-343.7
6	0.0	609.7	455.7	153.98	53.13	100.8	0.0	154.0	64.4	-279.3
7	0.0	609.7	455.7	153.98	53.13	100.8	10.1	143.9	52.0	-227.3
8	0.0	609.7	455.7	153.98	26.56	127.4	10.1	143.9	45.0	-182.3
9	0.0	609.7	455.7	154.0	0.00	154.0	12.7	141.2	38.2	-144.2
10	0.0	609.7	455.7	154.0	0.00	154.0	15.4	138.6	32.4	-111.8
11	0.0	609.7	455.7	154.0	0.00	154.0	15.4	138.6	28.0	-83.8
12	0.0	609.7	455.7	154.0	0.00	154.0	15.4	138.6	24.2	-59.6
13	0.0	609.7	455.7	154.0	0.00	154.0	15.4	138.6	20.9	-38.7
14	0.0	609.7	455.7	154.0	0.00	154.0	15.4	138.6	18.1	-20.6
15	0.0	609.7	455.7	154.0	0.00	154.0	15.4	138.6	15.7	-4.9
16	0.0	609.7	455.7	154.0	0.00	154.0	15.4	138.6	13.5	8.6
17	0.0	609.7	455.7	154.0	0.00	154.0	15.4	138.6	11.7	20.3
18	0.0	609.7	455.7	154.0	0.00	154.0	15.4	138.6	10.1	30.4
19	0.0	609.7	455.7	154.0	0.00	154.0	15.4	138.6	8.7	39.2
20	-24.2	609.7	455.7	154.0	0.00	154.0	15.4	162.8	8.9	48.1

Table 19: 7 Year NPV

Project year	Cap Ex	Revenue	CCOP	Gr. Profit	Deprcn	Taxbl Inc	Tax Paid	Cash Flow	PV of CF	NPV
1	138.4	0.0	0.0	0.00	0.00	0.0	0.0	-138.4	-119.6	-119.6
2	322.8	0.0	0.0	0.00	0.00	0.0	0.0	-322.8	-241.4	-361.0
3	0.0	0.0	0.0	0.00	65.90	-65.9	0.0	0.0	0.0	-361.0
4	0.0	0.0	0.0	0.00	112.94	-112.9	0.0	0.0	0.0	-361.0
5	24.2	304.8	244.7	60.11	80.66	-20.6	0.0	35.9	17.3	-343.7
6	0.0	609.7	455.7	153.98	57.60	96.4	0.0	154.0	64.4	-279.3
7	0.0	609.7	455.7	153.98	41.18	112.8	9.6	144.3	52.2	-227.1
8	0.0	609.7	455.7	153.98	41.14	112.8	11.3	142.7	44.6	-182.6
9	0.0	609.7	455.7	154.0	41.18	112.8	11.3	142.7	38.6	-144.0
10	0.0	609.7	455.7	154.0	20.57	133.4	11.3	142.7	33.3	-110.7
11	0.0	609.7	455.7	154.0	0.00	154.0	13.3	140.6	28.4	-82.2
12	0.0	609.7	455.7	154.0	0.00	154.0	15.4	138.6	24.2	-58.0
13	0.0	609.7	455.7	154.0	0.00	154.0	15.4	138.6	20.9	-37.1
14	0.0	609.7	455.7	154.0	0.00	154.0	15.4	138.6	18.1	-19.0
15	0.0	609.7	455.7	154.0	0.00	154.0	15.4	138.6	15.7	-3.4
16	0.0	609.7	455.7	154.0	0.00	154.0	15.4	138.6	13.5	10.2
17	0.0	609.7	455.7	154.0	0.00	154.0	15.4	138.6	11.7	21.9
18	0.0	609.7	455.7	154.0	0.00	154.0	15.4	138.6	10.1	32.0
19	0.0	609.7	455.7	154.0	0.00	154.0	15.4	138.6	8.7	40.7
20	-24.2	609.7	455.7	154.0	0.00	154.0	15.4	162.8	8.9	49.6

DCFROR:

This is known to be the highest the interest rate in which our MTO plant will be funded at, such that we will at least break even on our NPV value. To do this, using goal seek the NPV value at each depreciation period was set equal to 0 by changing the 15.65 original interest rate. The tables below show the maximum DCFROR at each depreciation period.

Table 20: 5 Year DCFROR

							Cost of Capital	17.38%		
Project year	Cap Ex	Revenue	CCOP	Gr. Profit	Deprcn	Taxbl Inc	Tax Paid	Cash Flow	PV of CF	NPV
1	138.4	0.0	0.0	0.0	0.0	0.0	0.0	-138.4	-117.9	-117.9
2	322.8	0.0	0.0	0.0	0.0	0.0	0.0	-322.8	-234.3	-352.2
3	0.0	0.0	0.0	0.0	92.2	-92.2	0.0	0.0	0.0	-352.2
4	0.0	0.0	0.0	0.0	147.6	-147.6	0.0	0.0	0.0	-352.2
5	24.2	304.8	244.7	60.1	88.5	-28.4	0.0	35.9	16.1	-336.1
6	0.0	609.7	455.7	154.0	53.1	100.8	0.0	154.0	58.9	-277.2
7	0.0	609.7	455.7	154.0	53.1	100.8	10.1	143.9	46.9	-230.3
8	0.0	609.7	455.7	154.0	26.6	127.4	10.1	143.9	39.9	-190.4
9	0.0	609.7	455.7	154.0	0.0	154.0	12.7	141.2	33.4	-157.1
10	0.0	609.7	455.7	154.0	0.0	154.0	15.4	138.6	27.9	-129.2
11	0.0	609.7	455.7	154.0	0.0	154.0	15.4	138.6	23.8	-105.4
12	0.0	609.7	455.7	154.0	0.0	154.0	15.4	138.6	20.2	-85.1
13	0.0	609.7	455.7	154.0	0.0	154.0	15.4	138.6	17.2	-67.9
14	0.0	609.7	455.7	154.0	0.0	154.0	15.4	138.6	14.7	-53.2
15	0.0	609.7	455.7	154.0	0.0	154.0	15.4	138.6	12.5	-40.7
16	0.0	609.7	455.7	154.0	0.0	154.0	15.4	138.6	10.7	-30.0
17	0.0	609.7	455.7	154.0	0.0	154.0	15.4	138.6	9.1	-20.9
18	0.0	609.7	455.7	154.0	0.0	154.0	15.4	138.6	7.7	-13.2
19	0.0	609.7	455.7	154.0	0.0	154.0	15.4	138.6	6.6	-6.6
20	-24.2	609.7	455.7	154.0	0.0	154.0	15.4	162.8	6.6	0.0

Table 21: 7 Year DCFROR

							Cost of Capital	17.44%		
Project year	Cap Ex	Revenue	CCOP	Gr. Profit	Deprcn	Taxbl Inc	Tax Paid	Cash Flow	PV of CF	NPV
1	138.4	0.0	0.0	0.0	0.0	0.0	0.0	-138.4	-117.8	-117.8
2	322.8	0.0	0.0	0.0	0.0	0.0	0.0	-322.8	-234.1	-351.9
3	0.0	0.0	0.0	0.0	65.9	-65.9	0.0	0.0	0.0	-351.9
4	0.0	0.0	0.0	0.0	112.9	-112.9	0.0	0.0	0.0	-351.9
5	24.2	304.8	244.7	60.1	80.7	-20.6	0.0	35.9	16.1	-335.8
6	0.0	609.7	455.7	154.0	57.6	96.4	0.0	154.0	58.7	-277.1
7	0.0	609.7	455.7	154.0	41.2	112.8	9.6	144.3	46.9	-230.3
8	0.0	609.7	455.7	154.0	41.1	112.8	11.3	142.7	39.4	-190.8
9	0.0	609.7	455.7	154.0	41.2	112.8	11.3	142.7	33.6	-157.3
10	0.0	609.7	455.7	154.0	20.6	133.4	11.3	142.7	28.6	-128.7
11	0.0	609.7	455.7	154.0	0.0	154.0	13.3	140.6	24.0	-104.7
12	0.0	609.7	455.7	154.0	0.0	154.0	15.4	138.6	20.1	-84.5
13	0.0	609.7	455.7	154.0	0.0	154.0	15.4	138.6	17.1	-67.4
14	0.0	609.7	455.7	154.0	0.0	154.0	15.4	138.6	14.6	-52.8
15	0.0	609.7	455.7	154.0	0.0	154.0	15.4	138.6	12.4	-40.4
16	0.0	609.7	455.7	154.0	0.0	154.0	15.4	138.6	10.6	-29.8
17	0.0	609.7	455.7	154.0	0.0	154.0	15.4	138.6	9.0	-20.7
18	0.0	609.7	455.7	154.0	0.0	154.0	15.4	138.6	7.7	-13.1
19	0.0	609.7	455.7	154.0	0.0	154.0	15.4	138.6	6.5	-6.5
20	-24.2	609.7	455.7	154.0	0.0	154.0	15.4	162.8	6.5	0.0

Sensitivity Analysis:

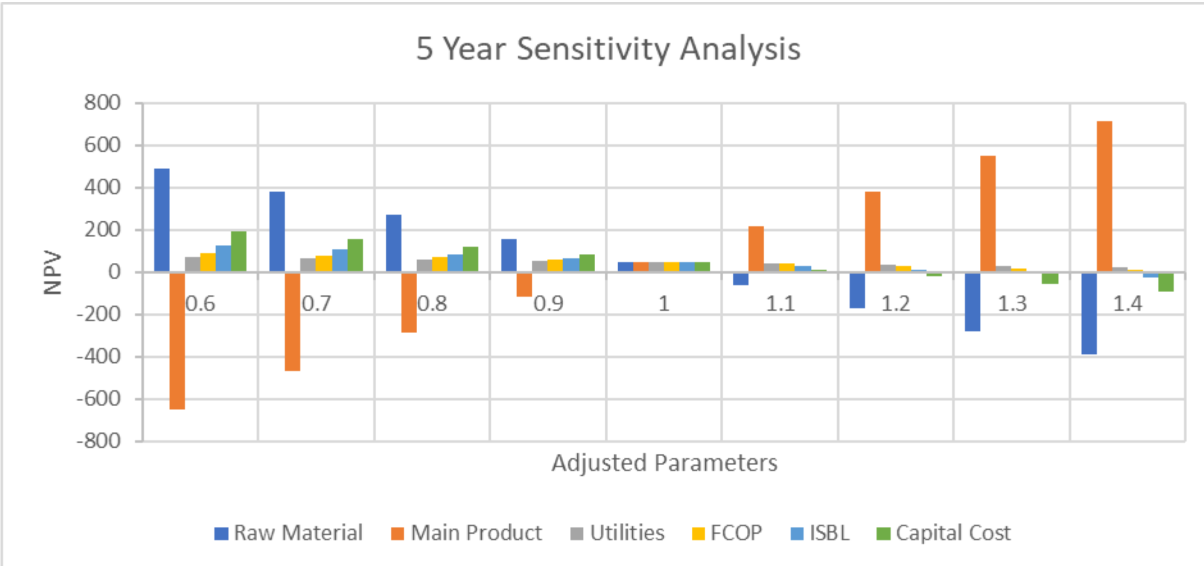

Figure 20: 5 Year Sensitivity Analysis

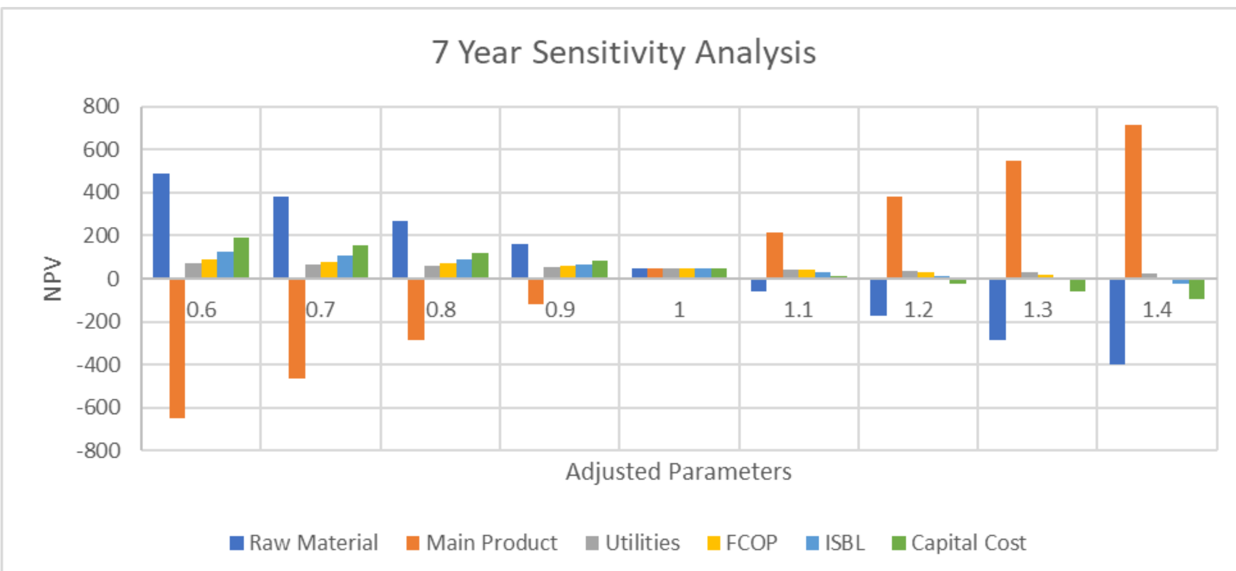

Figure 21: 7 Year Sensitivity Analysis

Discussion of Sensitivity Analysis:

The sensitivity analysis was done from the range of 0.6-1.4. Our value for each component was multiplied throughout this range and the NPV of our process was obtained at each specific point. We were able to identify how a change in each component can affect our total NPV value of the process. The trend which has become very clear as shown above in both depreciation periods, as our raw material cost increased there was a decrease in our NPV. Looking at our main product, as the revenue of our ethylene and propylene so did our NPV. Lastly looking at our utilities, FCOP, ISBL, and Capital cost there was a decrease in the NPV as these costs increased throughout our range. This means and shows how many factors can limit the NPV and what factors can help in increasing the NPV.

Appendix 1 – Design Basis

The basis of this process is to produce ethylene and propylene and their associated by-products to be able to meet the growing demand in India, specifically the west coast of India. Ethylene has a variety of uses which include the production of polyethylene to make plastics, as well as ethylene oxide and ethylene glycol to produce polyester fibers.[11] Due to the necessity of these petrochemicals, the increasing global consumption, and the increasing consumption in India alone, the desired number of olefins and their related by-products to produce is 667,384 total metric tons per year. The 667,384 metric tons of olefins include 99.4% purity ethylene and 99.4% purity propylene. To meet this goal, approximately 1,623,000 metric tons of methanol will be required per year. The feed stream only consists of methanol. The catalyst chosen is the SAPO-34 catalyst, and the process will require approximately 39.6 metric tons of this catalyst per year. With these raw materials, the desired production rate of olefins per year will be met. The product compositions include 99.4% pure ethylene, with the remaining 0.06% consisting of hydrogen, carbon monoxide, carbon dioxide, methane, ethane, propylene, propane, butylene, and pentane. The propylene product stream consists of 99.4% propylene, with the remaining 0.06% consisting of carbon dioxide, methane, ethylene, ethane, propane, butylene and pentene.

Appendix 2 – BFD

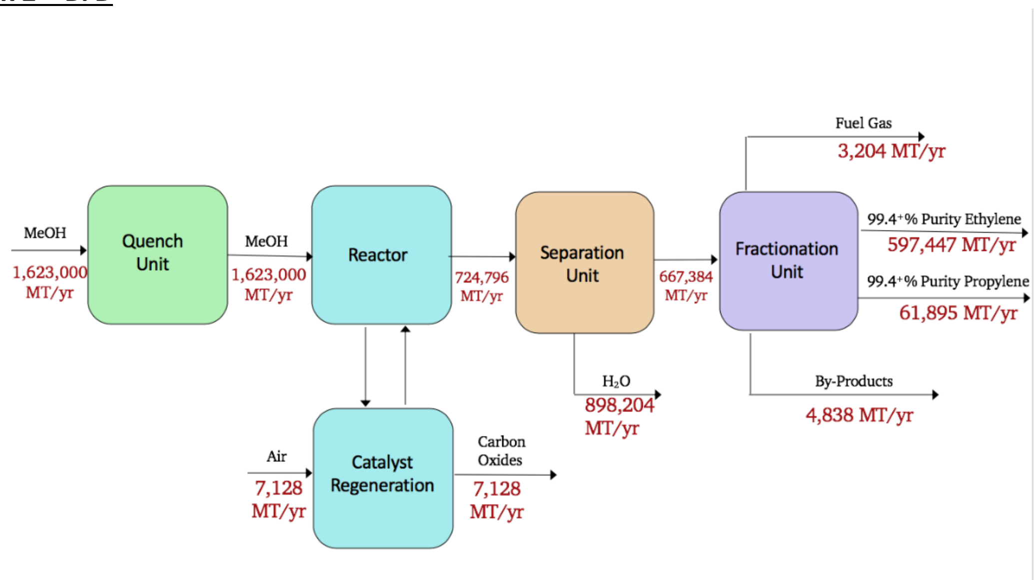

Figure 22: Simplified BFD of UOP's Advanced MTO Process

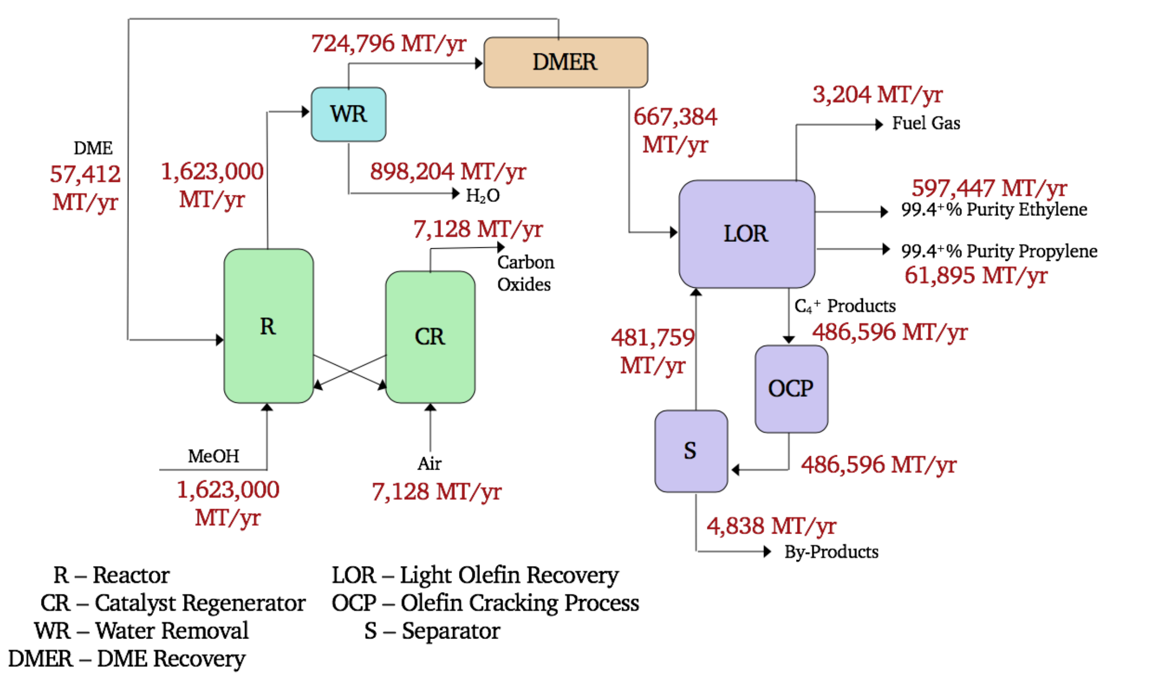

Figure 23: BFD of UOP's Advanced MTO Process [12]

Operating conditions, Flow rates, and Assumptions

1. The production rate of olefins is 667,384 MT/yr.
2. The reactor operates in vapor phase, a pressure 2 bar, and at a temperature of 450 °C.[7]
3. Assume a 100% conversion of methanol in the reactor.[12]
4. Reactor is a fluidized bed reactor.[13]
5. The selectivity's of the carbon species are known.[7]
6. Assume 90% MeOH conversion to olefins.[12]
7. The weight hour space velocity (WHSV) of the SAPO-34 catalyst is 1 hour.
8. Average coke content is 8%.[13]
9. Assume 100% of the water is removed.
10. Assume 2% CO_2 is released into the environment.
11. Assume 90% selectivity DME in the separator.
12. Assume minimal trace amounts of DME are left in the stream after the removal stage.
13. Assume polymer grade (99.4% pure) ethylene and propylene is produced.

Appendix 3 – Final PFD

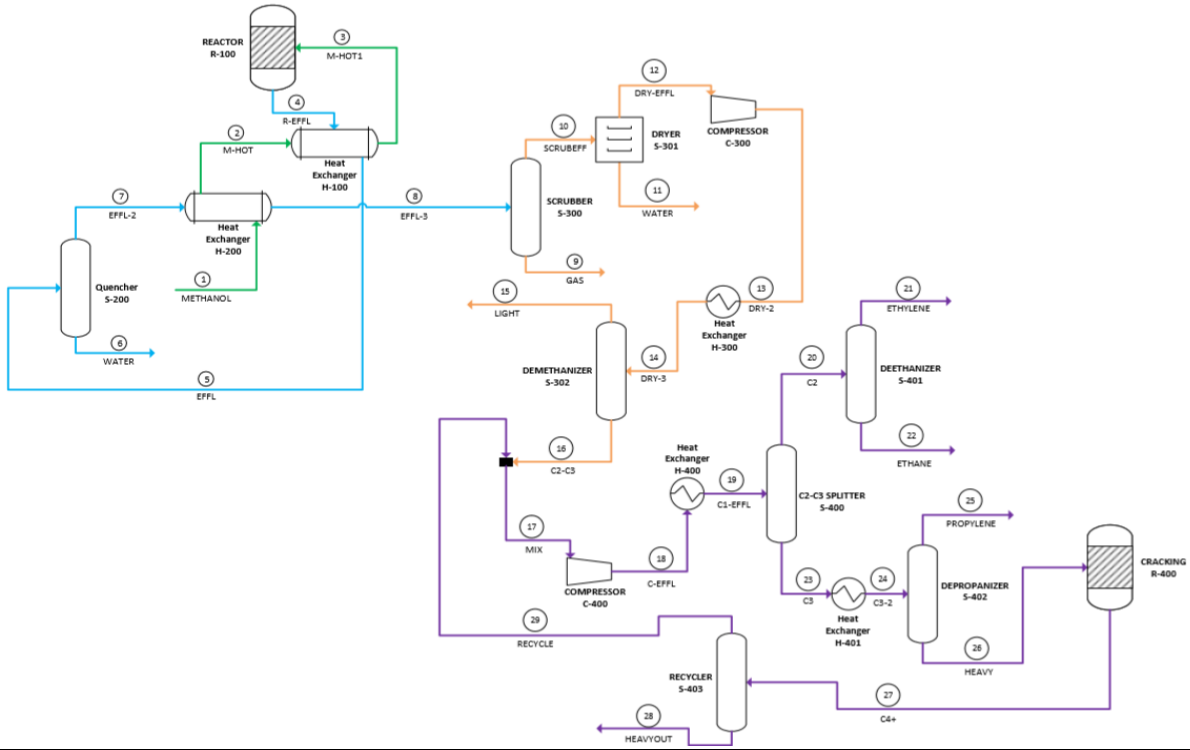

Figure 24: Advanced MTO Preliminary PFD [10]

Appendix 4 – Material/Mass Balances and Aspen Simulation

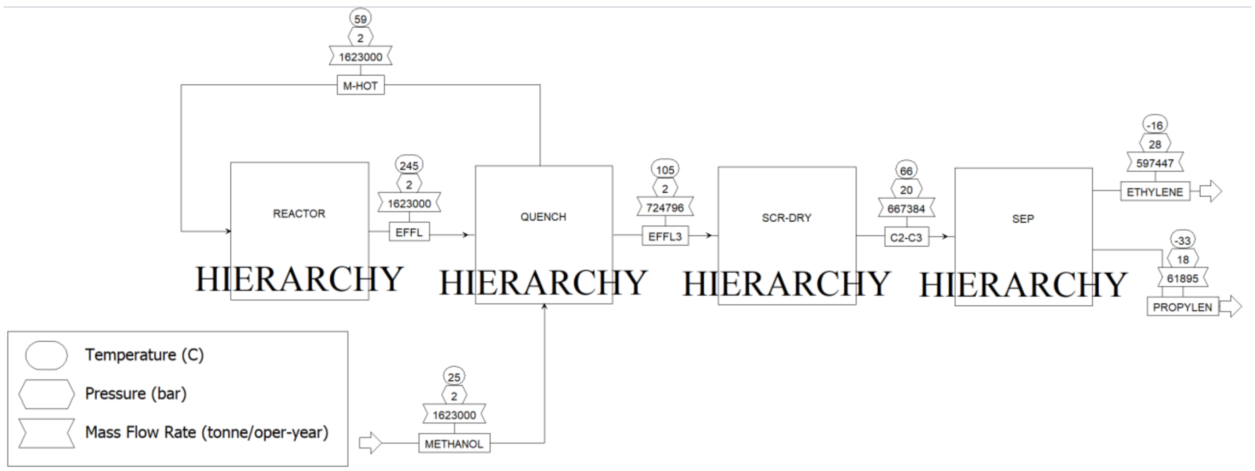

Figure 25: Aspen MTO PFD

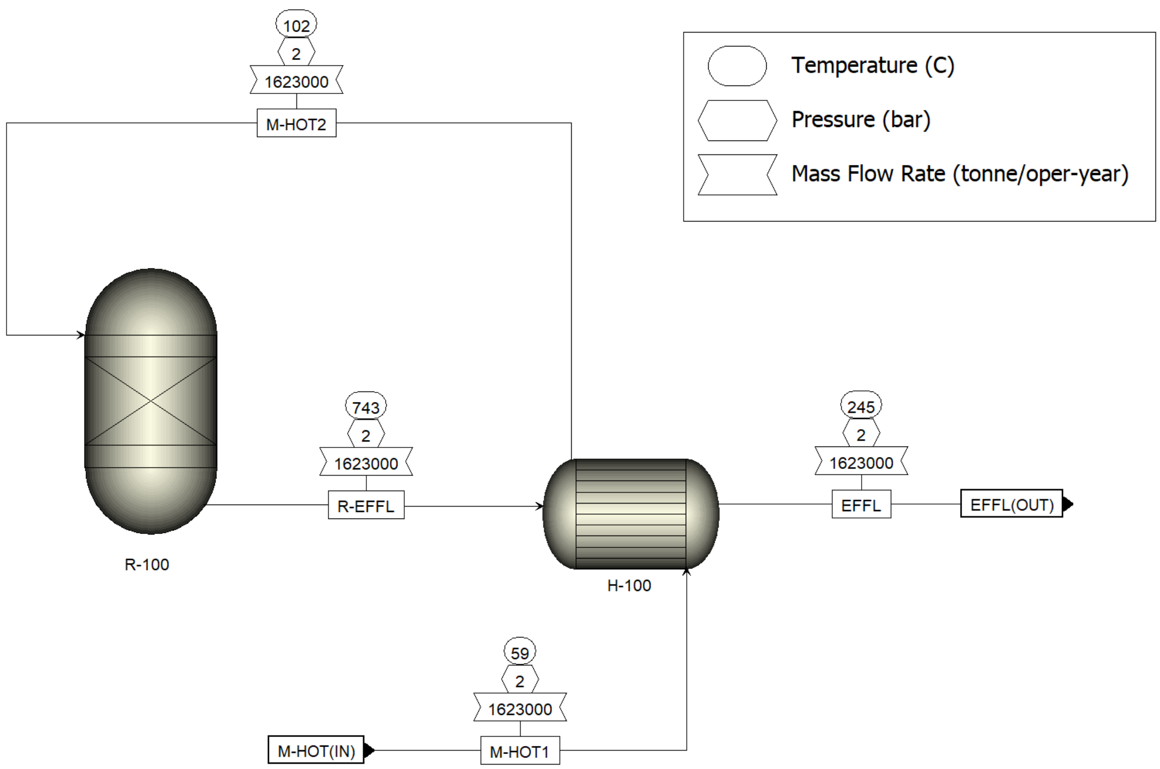

Figure 26: Reactor Hierarchy PFD

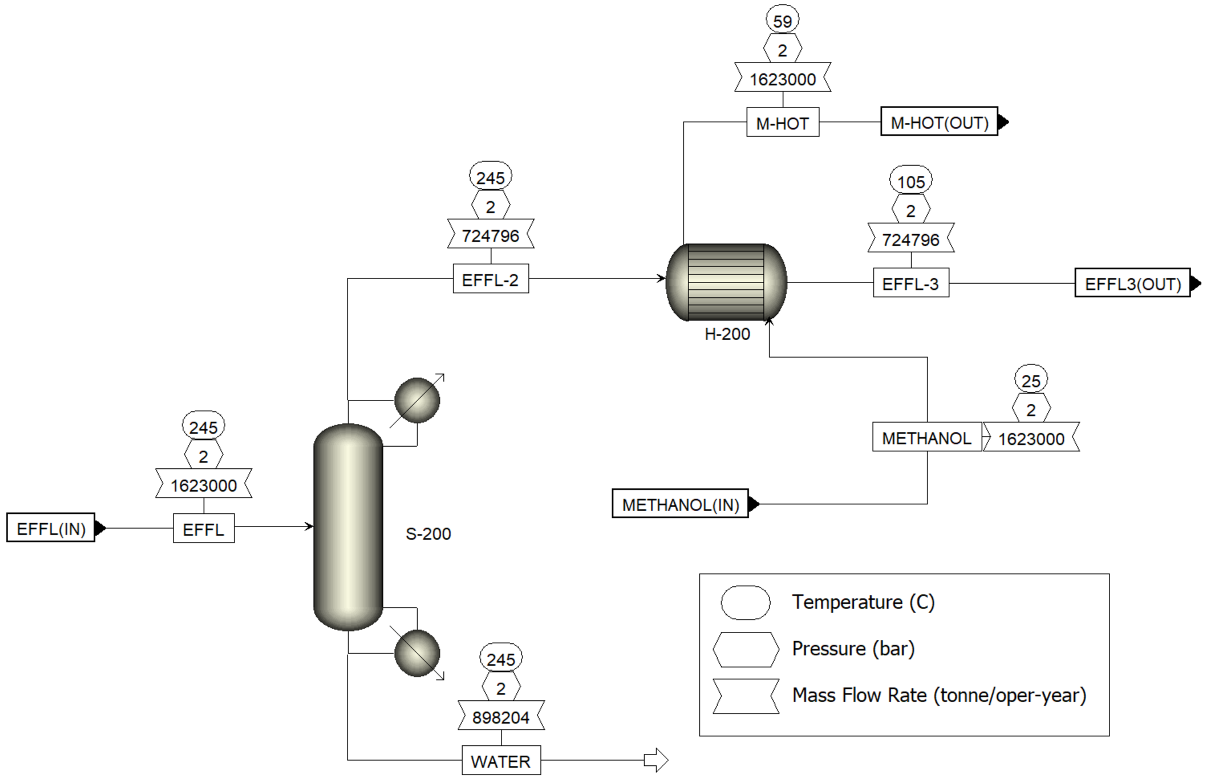

Figure 27: Quench Hierarchy PFD

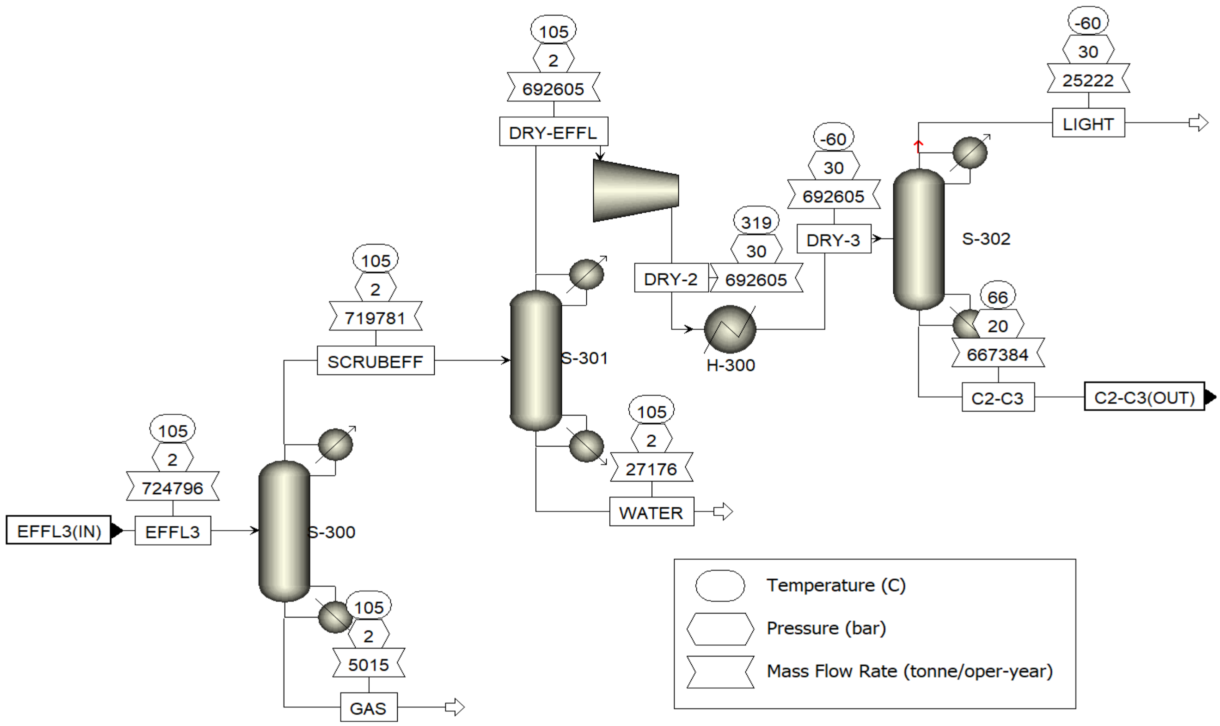

Figure 28: Scrubber/Dryer Hierarchy

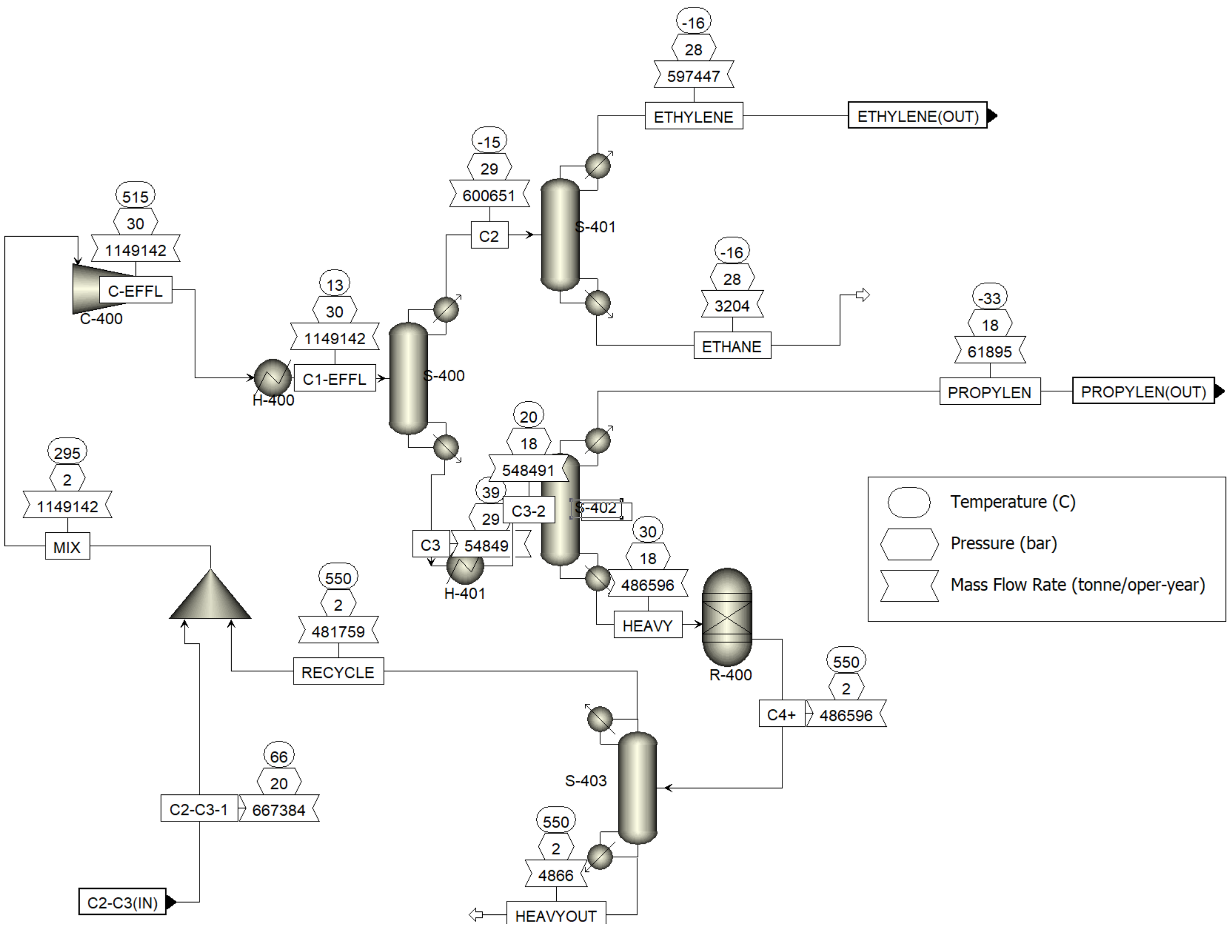

Figure 29: Sep PFD

Mass Balances on every part of the process:

Table 22: Material/Mass Balance Reactor Hierarchy

Stream Name	Units	METHANOL	M-HOT	EFFL	WATER	EFFL-2	EFFL-3
Description							
From			H-200	H-100	S-200	S-200	H-200
To		H-200	H-100	S-200		H-200	S-300
Phase		Liquid	Liquid	Vapor	Vapor	Vapor	Vapor
Temperature	C	25	59.7037	245	245	245	105.3
Pressure	bar	2.2	2.2	2	2	2	2
Molar Vapor Fraction		0	0	1	1	1	1
Molar Liquid Fraction		1	1	0	0	0	0
HYDROGEN	kmol/hr	0	0	1.62113	0	1.62113	1.62113
CO	kmol/hr	0	0	16.0853	0	16.0853	16.0853
CO2	kmol/hr	0	0	13.1568	0	13.1568	13.1568
METHANE	kmol/hr	0	0	153.237	0	153.237	153.237
ETHYLENE	kmol/hr	0	0	1432.85	0	1432.85	1432.85
ETHANE	kmol/hr	0	0	14.7014	0	14.7014	14.7014
PROPYLEN	kmol/hr	0	0	623.824	0	623.824	623.824
PROPANE	kmol/hr	0	0	7.35071	0	7.35071	7.35071
BUTYLENE	kmol/hr	0	0	146.209	0	146.209	146.209
PENTENE	kmol/hr	0	0	23.3934	0	23.3934	23.3934
METHANOL	kmol/hr	5778.24	5778.24	0	0	0	0
WATER	kmol/hr	0	0	5806.07	5631.9	174.176	174.176
Total Mole Flows	kmol/hr	5778.24	5778.24	8238.5	5631.9	2606.6	2606.6
HYDROGEN	kg/hr	0	0	3.268	0	3.268	3.268
CO	kg/hr	0	0	450.556	0	450.556	450.556
CO2	kg/hr	0	0	579.027	0	579.027	579.027
METHANE	kg/hr	0	0	2458.35	0	2458.35	2458.35
ETHYLENE	kg/hr	0	0	40196.7	0	40196.7	40196.7
ETHANE	kg/hr	0	0	442.067	0	442.067	442.067
PROPYLEN	kg/hr	0	0	26250.9	0	26250.9	26250.9
PROPANE	kg/hr	0	0	324.141	0	324.141	324.141
BUTYLENE	kg/hr	0	0	8203.41	0	8203.41	8203.41
PENTENE	kg/hr	0	0	1640.68	0	1640.68	1640.68
METHANOL	kg/hr	185147	185147	0	0	0	0
WATER	kg/hr	0	0	104598	101460	3137.84	3137.84
Total Mass Flows	kg/hr	185147	185147	185147	101460	83686.9	83686.9

Table 23: Material/Mass Balance Quench Hierarchy

Stream Name	Units	M-HOT	M-HOT2	R-EFFL	EFFL
From		H-200	H-100	R-100	H-100
To		H-100	R-100	H-100	S-200
Phase		Liquid	Mixed	Vapor	Vapor
Temperature	C	59.7037	85.578	743.15	245
Pressure	bar	2.2	2.2	2	2
Molar Vapor Fraction		0	0.992869	1	1
Molar Liquid Fraction		1	0.007131	0	0
HYDROGEN	kmol/hr	0	0	1.62113	1.62113
CO	kmol/hr	0	0	16.0853	16.0853
CO2	kmol/hr	0	0	13.1568	13.1568
METHANE	kmol/hr	0	0	153.237	153.237
ETHYLENE	kmol/hr	0	0	1432.85	1432.85
ETHANE	kmol/hr	0	0	14.7014	14.7014
PROPYLEN	kmol/hr	0	0	623.824	623.824
PROPANE	kmol/hr	0	0	7.35071	7.35071
BUTYLENE	kmol/hr	0	0	146.209	146.209
PENTENE	kmol/hr	0	0	23.3934	23.3934
METHANOL	kmol/hr	5778.24	5778.24	0	0
WATER	kmol/hr	0	0	5806.07	5806.07
Total Mole Flows	kmol/hr	5778.24	5778.24	8238.5	8238.5
HYDROGEN	kg/hr	0	0	3.268	3.268
CO	kg/hr	0	0	450.556	450.556
CO2	kg/hr	0	0	579.027	579.027
METHANE	kg/hr	0	0	2458.35	2458.35
ETHYLENE	kg/hr	0	0	40196.7	40196.7
ETHANE	kg/hr	0	0	442.067	442.067
PROPYLEN	kg/hr	0	0	26250.9	26250.9
PROPANE	kg/hr	0	0	324.141	324.141
BUTYLENE	kg/hr	0	0	8203.41	8203.41
PENTENE	kg/hr	0	0	1640.68	1640.68
METHANOL	kg/hr	185147	185147	0	0
WATER	kg/hr	0	0	104598	104598
Total Mass Flows	kg/hr	185147	185147	185147	185147

Table 24: Material/Mass Balance Scrubber/Dryer Hierarchy

Stream Name	Units	EFFL-3	GAS	SCRUBEFF	WATER-DR	DRY-EFFL	DRY-2	DRY-3	C2-C3	LIGHT
Description										
From		H-200	S-300	S-300	S-301	S-301	C-300	H-300	S-302	S-302
To		S-300		S-301		C-300	H-300	S-302	B10	
MIXED Substream										
Phase		Vapor	Vapor	Vapor	Liquid	Vapor	Vapor	Liquid	Vapor	Vapor
Temperature	C	105.3	105.3	105.3	105.3	105.3	298.296	-60	66	-60
Pressure	bar	2	2	2	2	2	30.1	30	20	30
Molar Vapor Fraction		1	1	1	0	1	1	0	1	1
Molar Liquid Fraction		0	0	0	1	0	0	1	0	0
HYDROGEN	kmol/hr	1.62113	0	1.62113	0	1.62113	1.62113	1.62113	0	1.62113
CO	kmol/hr	16.0853	0	16.0853	0	16.0853	16.0853	16.0853	0	16.0853
CO2	kmol/hr	13.1568	13.1568	0	0	0	0	0	0	0
METHANE	kmol/hr	153.237	0	153.237	0	153.237	153.237	153.237	0	153.237
ETHYLENE	kmol/hr	1432.85	0	1432.85	0	1432.85	1432.85	1432.85	1432.85	0
ETHANE	kmol/hr	14.7014	0	14.7014	0	14.7014	14.7014	14.7014	14.7014	0
PROPYLEN	kmol/hr	623.824	0	623.824	0	623.824	623.824	623.824	623.824	0
PROPANE	kmol/hr	7.35071	0	7.35071	0	7.35071	7.35071	7.35071	7.35071	0
BUTYLENE	kmol/hr	146.209	0	146.209	0	146.209	146.209	146.209	146.209	0
PENTENE	kmol/hr	23.3934	0	23.3934	0	23.3934	23.3934	23.3934	23.3934	0
METHANOL	kmol/hr	0	0	0	0	0	0	0	0	0
WATER	kmol/hr	174.176	0	174.176	174.176	0	0	0	0	0
Total Mole Flows	kmol/hr	2606.6	13.1568	2593.44	174.176	2419.27	2419.27	2419.27	2248.32	170.944
HYDROGEN	kg/hr	3.268	0	3.268	0	3.268	3.268	3.268	0	3.268
CO	kg/hr	450.556	0	450.556	0	450.556	450.556	450.556	0	450.556
CO2	kg/hr	579.027	579.027	0	0	0	0	0	0	0
METHANE	kg/hr	2458.35	0	2458.35	0	2458.35	2458.35	2458.35	0	2458.35
ETHYLENE	kg/hr	40196.7	0	40196.7	0	40196.7	40196.7	40196.7	40196.7	0
ETHANE	kg/hr	442.067	0	442.067	0	442.067	442.067	442.067	442.067	0
PROPYLEN	kg/hr	26250.9	0	26250.9	0	26250.9	26250.9	26250.9	26250.9	0
PROPANE	kg/hr	324.141	0	324.141	0	324.141	324.141	324.141	324.141	0
BUTYLENE	kg/hr	8203.41	0	8203.41	0	8203.41	8203.41	8203.41	8203.41	0
PENTENE	kg/hr	1640.68	0	1640.68	0	1640.68	1640.68	1640.68	1640.68	0
METHANOL	kg/hr	0	0	0	0	0	0	0	0	0
WATER	kg/hr	3137.84	0	3137.84	3137.84	0	0	0	0	0
Total Mass Flows	kg/hr	83686.9	579.027	83107.9	3137.84	79970.1	79970.1	79970.1	77057.9	2912.17

Table 25: Material/Mass Balance Sep Hierarchy

Stream Name	Units	C2-C3	RECYCLE	MIX	C-EFFL	C1-EFFL	C2	ETHYLENE	ETHANE	C3	C3-2	PROPYLEN	HEAVY	C4+	HEAVYOUT
From		S-302	S-403	B10	C-400	H-400	S-400	S-401	S-401	S-400	H-401	S-402	S-402	R-400	S-403
To		B10	B10	C-400	H-400	S-400	S-401			H-401	S-402		R-400	S-403	
Phase		Vapor	Vapor	Vapor	Vapor	Mixed	Vapor	Vapor	Liquid	Liquid	Mixed	Vapor	Liquid	Vapor	Vapor
Temperature	C	66	550	295	515.304	13.4	-14.9531	-16.3654	-15.6311	39.2517	20	-32.9657	30.0228	550	550
Pressure	bar	20	2	2	30.1	30.1	29	28	28	29	18	18	18	2	2
Molar Vapor Fraction		1	1	1	1	0.557966	1	1	0	0	0.170775	1	0	1	1
Molar Liquid Fraction		0	0	0	0	0.442034	0	0	1	1	0.829225	0	1	0	0
HYDROGEN	kmol/hr	0	0	0	0	0	0	0	0	0	0	0	0	0	0
CO	kmol/hr	0	0	0	0	0	0	0	0	0	0	0	0	0	0
CO2	kmol/hr	0	0	0	0	0	0	0	0	0	0	0	0	0	0
METHANE	kmol/hr	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ETHYLENE	kmol/hr	1432.85	1470.51	2903.35	2903.35	2903.35	2463.63	2451.11	12.5213	439.723	439.723	254.874	184.848	1485.32	14.8532
ETHANE	kmol/hr	14.7014	0	14.7014	14.7014	14.7014	7.7442	7.16411	0.580081	6.95725	6.95725	0.00013038	6.95712	0	0
PROPYLEN	kmol/hr	623.824	341.368	965.192	965.192	965.192	0.0116081	0.0056557	0.0059524	965.181	965.181	3.24E-72	965.181	344.807	3.44807
PROPANE	kmol/hr	7.35071	0	7.35071	7.35071	7.35071	7.52E-06	2.73E-06	4.79E-06	7.3507	7.3507	3.44E-86	7.3507	0	0
BUTYLENE	kmol/hr	146.209	0	146.209	146.209	146.209	1.09E-10	2.03E-17	1.09E-10	146.209	146.209	6.75E-150	146.209	0	0
PENTENE	kmol/hr	23.3934	0	23.3934	23.3934	23.3934	1.11E-18	0	0	23.3934	23.3934	5.37E-227	23.3934	0	0
METHANOL	kmol/hr	0	0	0	0	0	0	0	0	0	0	0	0	0	0
WATER	kmol/hr	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Total Mole Flows	kmol/hr	2248.32	1811.88	4060.2	4060.2	4060.2	2471.39	2458.28	13.1073	1588.81	1588.81	254.874	1333.94	1830.13	18.3013
HYDROGEN	kg/hr	0	0	0	0	0	0	0	0	0	0	0	0	0	0
CO	kg/hr	0	0	0	0	0	0	0	0	0	0	0	0	0	0
CO2	kg/hr	0	0	0	0	0	0	0	0	0	0	0	0	0	0
METHANE	kg/hr	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ETHYLENE	kg/hr	40196.7	41253.3	81450	81450	81450	69114.2	68762.9	351.269	12335.9	12335.9	7150.18	5185.69	41668.8	416.688
ETHANE	kg/hr	442.067	0	442.067	442.067	442.067	232.865	215.422	17.4428	209.202	209.202	0.00392057	209.198	0	0
PROPYLEN	kg/hr	26250.9	14365	40615.9	40615.9	40615.9	0.488476	0.237994	0.250482	40615.4	40615.4	1.36E-70	40615.4	14509.7	145.097
PROPANE	kg/hr	324.141	0	324.141	324.141	324.141	0.0003317	0.0001204	0.0002113	324.14	324.14	1.51E-84	324.14	0	0
BUTYLENE	kg/hr	8203.41	0	8203.41	8203.41	8203.41	6.13E-09	1.14E-15	6.13E-09	8203.41	8203.41	3.79E-148	8203.41	0	0
PENTENE	kg/hr	1640.68	0	1640.68	1640.68	1640.68	7.81E-17	0	0	1640.68	1640.68	3.76E-225	1640.68	0	0
METHANOL	kg/hr	0	0	0	0	0	0	0	0	0	0	0	0	0	0
WATER	kg/hr	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Total Mass Flows	kg/hr	77057.9	55618.3	132676	132676	132676	69347.5	68978.5	368.962	63328.7	63328.7	7150.19	56178.5	56178.5	561.785

Table 26: Sep Hierarchy Equipment List

SEP Hiearchy Equipment					
1	R-400	R-Yield Reactor	40991.59	823.15 K, 2 bar	Carbon Steel
Reboilers					
#	**Aspen ID**	**Industry Name**	**Duty (kW)**	**Specifications**	**Material**
2	S-400	C2-C3 - Kettle Reboiler	8000	38 trays, 19 bar, 312.4 K	Carbon Steel
3	S-401	Deethanizer - Kettle Reboiler	114.52	60 trays, 28 bar, 272.52 K	Carbon Steel
4	S-402	Depropanizer - Kettle Reboiler	2000	195 trays, 18 bar, 303.17 K	Carbon Steel
Condensers					
#	**Aspen ID**	**Industry Name**	**Duty (kW)**	**Specifications**	**Material**
5	S-400	C2-C3 - Partial Vapor	-8000	38 trays, 19 bar, 258.25 K	Carbon Steel
6	S-401	Deethanizer - Partial Vapor	-1120.6	60 trays, 28 bar, 256.79 K	Carbon Steel
7	H-400	Shell&Tube Heat Exchanger	-50300	Inlet: Thot=788 K Outlet: Thot=287 K	Carbon Steel
8	H-401	Shell&Tube Heat Exchanger	-23005	Inlet: Thot=312 K Outlet: Thot=293 K	Carbon Steel
9	S-402	Depropanizer - Partial Vapor	-2000	195 trays, 18 bar, 240.18 K	Carbon Steel
Compressor					
#	**Aspen ID**	**Industry Name**	**Duty (kW)**	**Specifications**	**Material**
10	C-400	Isentropic Compressor	22108	30.1 bar, 465.21 K	Carbon Steel

Table 27: Reactor Hierarchy Equipment List

REACTOR Hiearchy Equipment					
#	Aspen ID	Industry Name	Duty (kW)	Specifications	Material
1	R-100	R-Yield Reactor	56245.6	1016.3 K, 2 bar	Carbon Steel
2	H-100	Shell&Tube Heat Exchanger	38146.3	Inlet: Thot=387.25 K Tcold=298.15 K Outlet: Thot=378.45 K Tcold=300.139 K	Carbon Steel

Table 28: Quench Hierarchy Equipment List

QUENCH Hiearchy Equipment					
#	Aspen ID	Industry Name	Duty (kW)	Specifications	Material
1	H-100	Shell&Tube Heat Exchanger	546.578	Inlet: Thot=387.25 K Tcold=298.15 K Outlet: Thot=378.45 K Tcold=300.139 K	Carbon Steel

Table 29: Scrubber/Dryer Hierarchy Equipment List

SCR/DRY Hiearchy Equipment					
#	**Aspen ID**	**Industry Name**	**Duty (kW)**	**Specifications**	**Material**
1	C-300	Isentropic Compressor	56245.6	1016.3 K, 2 bar	Carbon Steel
2	H-300	Shell&Tube Heat Exchanger	38146.3	Inlet: Thot=387.25 K Tcold=298.15 K Outlet: Thot=378.45 K Tcold=300.139 K	Carbon Steel
3	S-302	Demethanizer - Kettle Reboiler	310.1	34 trays, 30 bar, 255.78 K	Carbon Steel

Appendix V

Hand Calculations

The sizing of one of the distillation columns, the deethanizer, in the process was calculated by hand. The values obtained are compared to the results obtained from aspen below.

The first step to develop a McCabe-Thiele diagram for the separation of ethylene and ethane. The diagram is shown below in Figure 21.

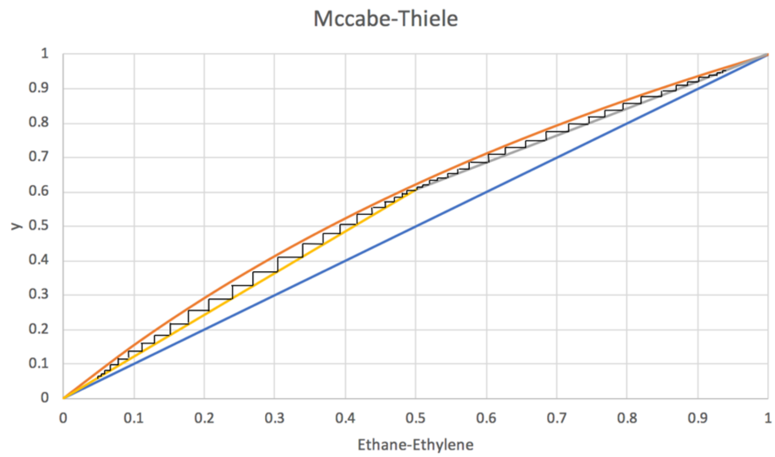

Figure 30: McCabe-Thiele Diagram for Ethylene and Ethane

The number of steps on the diagram corresponds to the number of theoretical stages. The McCabe-Thiele diagram suggests that there are 47 stages in the process.

The number of stages was also calculated using the Fenske Equation

$$N = \frac{log[(\frac{x_d}{1-x_d})(\frac{1-x_b}{x_b})]}{log(\propto)}$$

$$N = \frac{log[(\frac{0.738348}{1-0.738348})(\frac{1-(1.6326 \cdot 10^{-13})}{1.6326 \cdot 10^{-13}})]}{log(1.647)}$$

$$N = 61.1$$

Therefore, the number of trays required to achieve separation of ethylene and ethane is 62.

To determine the diameter of the column, the maximum allowable superficial vapor velocity was calculated.

$$\hat{u}_v = (-0.171 l_t^2 + 0.27 l_t - 0.047)[\frac{\rho_L - \rho_V}{\rho_V}]^{1/2}$$

Assuming the tray spacing is 0.6096 m, then

$$\hat{u}_v = (-0.171 \cdot (0.6096)^2 + 0.27 \cdot (0.6096) - 0.047)[\frac{(6.09143 - 1.64156)\frac{kmol}{m^3}}{1.64156\ \frac{kmol}{m^3}}]^{1/2}$$

$$\hat{u}_v = 0.0889\ \frac{m}{s}$$

The diameter of the column can then be calculated using the following equation

$$D_c = \sqrt{\frac{4\hat{V}_W}{\pi \rho_V \hat{u}_v}}$$

$$D_c = \sqrt{\frac{4 \cdot (0.645\frac{kmol}{s})}{\pi(1.64\frac{kmol}{m^3})(0.0889\ \frac{m}{s})}}$$

$$D_c = 2.37\ m$$

Therefore, the diameter of the column will be 2.37 meters (7.77 ft).

The height of the distillation column is determined by multiplying the number of actual stages by the tray spacing.

$$62\ stages \cdot 0.6096\ m = 37.8\ m$$

Assuming that extra space will be needed at the top and bottom of the distillation column, 15% of the height determined by the number of trays will be added

$$37.8\ m \cdot\ 0.15 = 5.67m$$
$$37.8\ m + 5.67\ m = 43.5\ m$$

Therefore, the height of the deethanizer will be 43.5 m (142.7 ft).

Aspen comparison

Table 30: Comparison of hand calculations vs. aspen calculations

Variable	Hand	Aspen
Stages	62	60
Height (ft)	142.7	178
Diameter (ft)	7.77	5.5

The required number of stages calculated by hand is 62 stages, while aspen determined the number of stages to be 60. There is a 3% error in the difference between the hand calculations and the aspen simulation data. The number of stages determined by the McCabe-Thiele diagram is less accurate, which can be due to errors in determining the operating lines. The hand calculations show a column height of 142.7 ft, while the aspen simulation shows a height of 178 ft for the deethanizer. The difference can be explained due to the amount of extra space that was added in the hand calculations. There is a 19.8% error in the difference between the hand calculations and the aspen simulation data for the height of the column. The hand calculations show a diameter of 7.77 ft, while aspen shows a diameter of 5 ft for the deethanizer column. Lastly there is 40% error in the difference between the hand calculations and the aspen simulation data for the diameter of the column.

Appendix VI: Utilities

Table 31: Utility Usage for MTO

Name	Type	Usage	Units
Electricity	Power	78,353.6	**kW**
Cooling Water	Water	8,924.7	**MT/hr**
R1270 (Propylene)	Refrigerant	42,730.9	**MT/hr**
Ethylene	Refrigerant	5,586.4	**MT/hr**
Steam (261.1 psig to 493.1 psig)	High Pressure Steam	23.3	**MT/hr**

In the table above it shows the major types of utilities that will be used in the MTO process. Three of the utilities will be coming from other resources, those three utilities being electricity, steam and cooling water, while excess propylene from the process will be used as the refrigerant, or also known as R1270 as well as ethylene which is known as R1150.

The first of the utilities is electricity. Electricity is a very important aspect of the MTO plant because it will be used in most if not all aspects of the plant. The most important places being in the third and fourth hierarchies, SCR-DRY and SEP. In both the SCR-DRY and SEP hierarchies there is an isentropic compressor that will have power being supplied to it. The isentropic compressor in SCR-DRY, C-300, takes the DRY-EFFL, the top product stream of S-301, and compresses it from 2 bar to 30 bar, which in turn causes the temperature to increase from 105 C to 319 C. The isentropic compressor in the SEP hierarchy, C-400, takes the MIX stream, which is a mixture of the recycle stream from S-403 and the C2-C3(IN) stream, and compresses it from 2 bar to 30 bar, which also causes a proportional rise in temperature going form 295 C to C-EFFL stream at 515 C.

The next utility is the cooling water, and that will be utilized in again the third and fourth hierarchies. This utility is used with either propylene or ethylene to further cool the streams. In the third hierarchy, the cooling water is used going from stream DRY-2 to DRY-3, which is the in and out streams of the Shell and Tube Heat Exchanger, H-300. H-300 takes the DRY-2 stream in and uses the cooling water and ethylene to decrease the temperature from 319 C to -60 C in stream DRY-3. In the fourth hierarchy, cooling water and propylene is used in the Shell and Tube Heat Exchanger H-400, taking C-EFFL as in the in stream and having C1-EFFL as the out stream. H-400 takes C-EFFL and cools it from 515 C to 13 C in C1-EFFL, which is then further cooled in S-400's top product stream through the condenser, C2, at -15 C with ethylene. The C2 stream is further cooled again, very slightly, to –16 C as the top product stream S-401, Ethylene. Another instance where cooling water is utilized is in H-401 where is takes C3 in, S-400's bottoms product stream, C3-2 as the exit stream. H-401 takes stream C3 and cools it to stream C3-2 going from 39 C to 20 C using ethylene. The stream is further cooled down to –33 C when going through S-402's condenser and exits as the top product stream, as propylene.

After that, the next utility is the refrigerant, R1270 and the refrigerant ethylene, which is the excess propylene from the product stream that will be used throughout the plant. The propylene will be kept in the gaseous state and will be distributed throughout the process where it is necessary. One of those places includes the storage of the actual product of ethylene and propylene, which need to be kept at a very low temperature.

The second to last utility to be talked about is steam, and that will be used in the reboilers of the distillation columns and compressors in hierarchies three and four, SCR-DRY and SEP respectively. The reboiler in SEP for S-400 requires high pressure steam to change the pressure, in turn affecting the temperature. This is seen when the C1-EFFL stream enters the separation unit at 30 bar and 13 C and exits at the bottom as C3 with a pressure of 29 bar and temperature of 39 C. The second instance where steam is used is when the C3-2 stream enters S-402 at 18 bar and 20 C and exits at the bottom as HEAVY with a pressure of 18 bar and temperature of 30 C, which requires high pressure steam. The first instance for the compressor, C-300, is the same as that for electricity also using high pressure steam to get the pressure to increase from 2 bar to 30 bar, which caused a major temperature to change from 105 C to 319 C. The same can be said for the second compressor, C-400, to also change the pressure from 2 bar to 30 bar, which also caused a major temperature change going from 295 C to 515 C.

The last utilities are air and nitrogen, and those will be used in the catalyst regeneration column and storage tanks, respectively. Air will be used in this vessel because it is necessary to burn off the coke on the catalyst. Nitrogen will be used in the storage tanks as a buffer because there could be run away gasses which could cause either a malfunction in the process or an explosion, both of which are bad. The utility total for nitrogen and air where not simulated and the values of the components were not estimated. For the sensitivity analysis report, this will be further investigated as we will be optimizing the size of the reactor, pressures, and temperatures of the equipment.

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